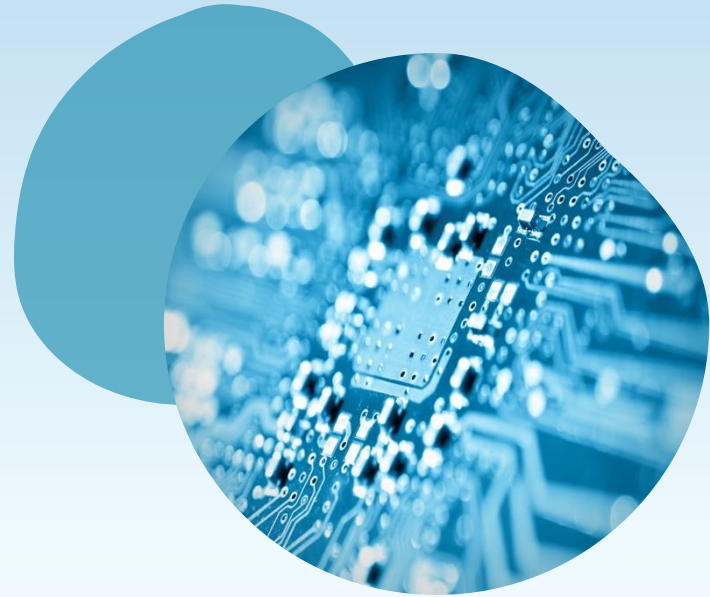


Introduction to TI MSP432 Launchpad

Embedded Software for the Internet Of Things

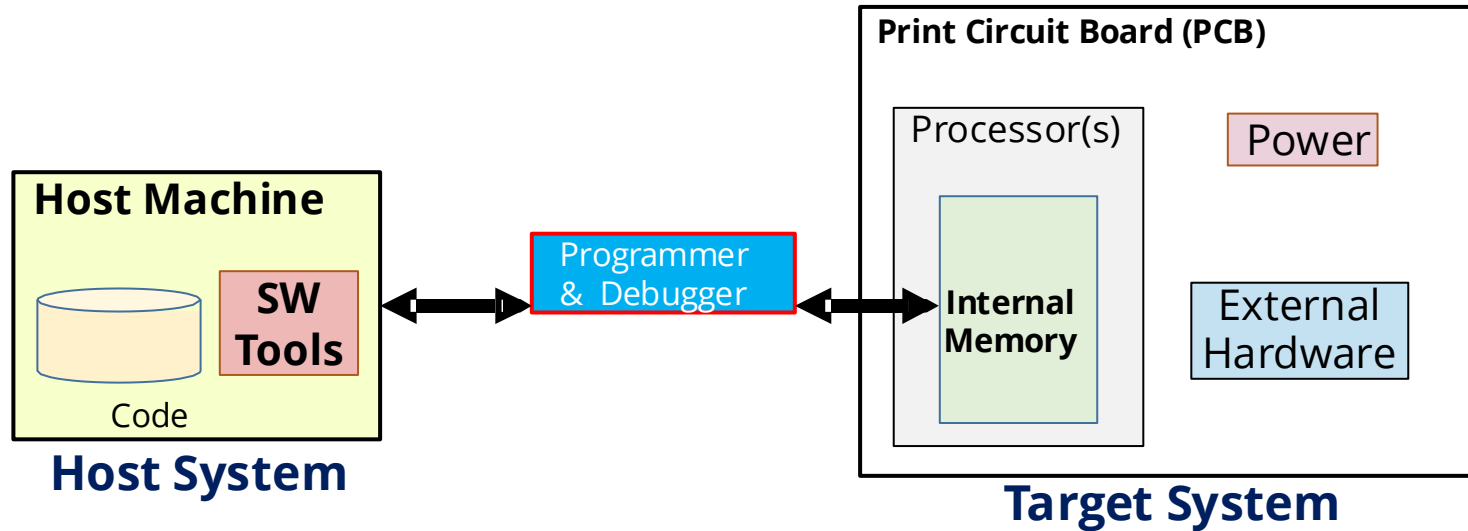
Kasim Sinan Yildirim



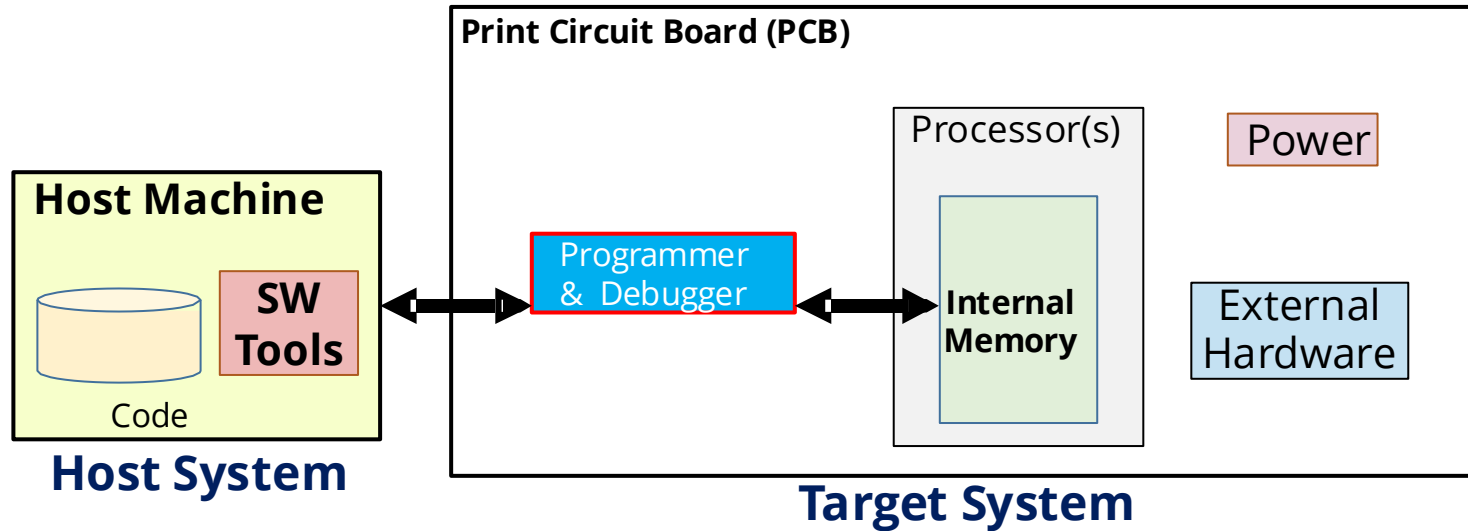
Embedded System Development Platform

- **Host system**
 - cross compiler, linker, and source-level debugger.
- **The target**
 - debugger, processor, sensors,...
- **Connections** between the host and the target system:
 - For downloading program images from the host system to the target system.
 - For transmitting **debugger information** between the host debugger and the target debug agent.

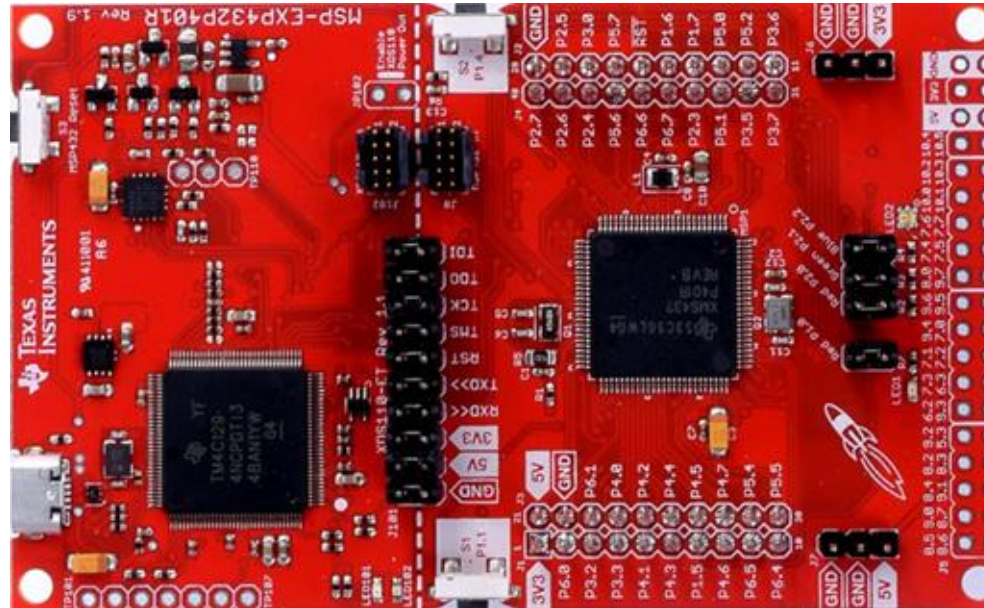
Embedded System Development Platform



Embedded System Development Platform

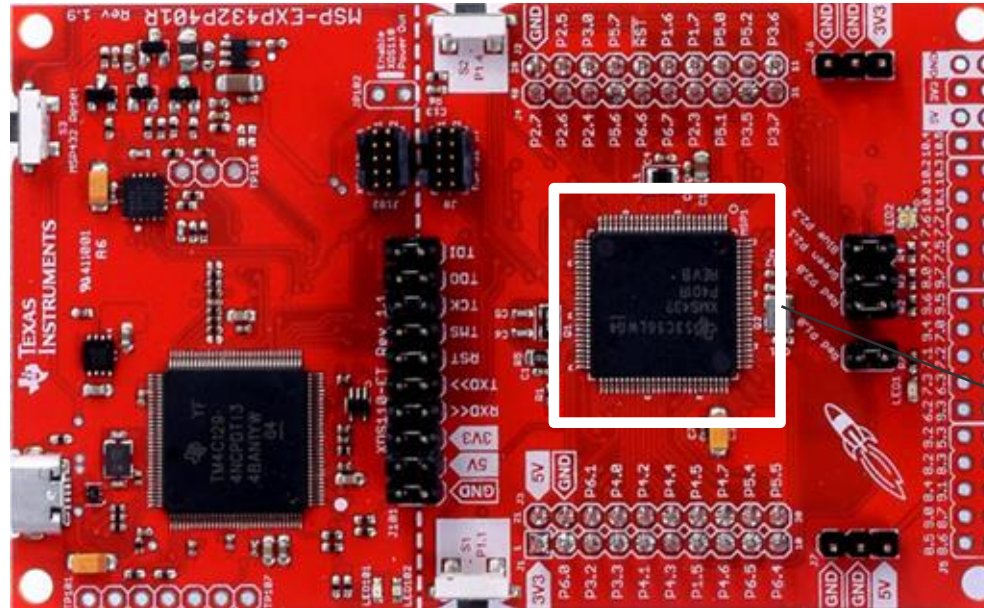


Embedded System Development Platform



Printed
Circuit
Board

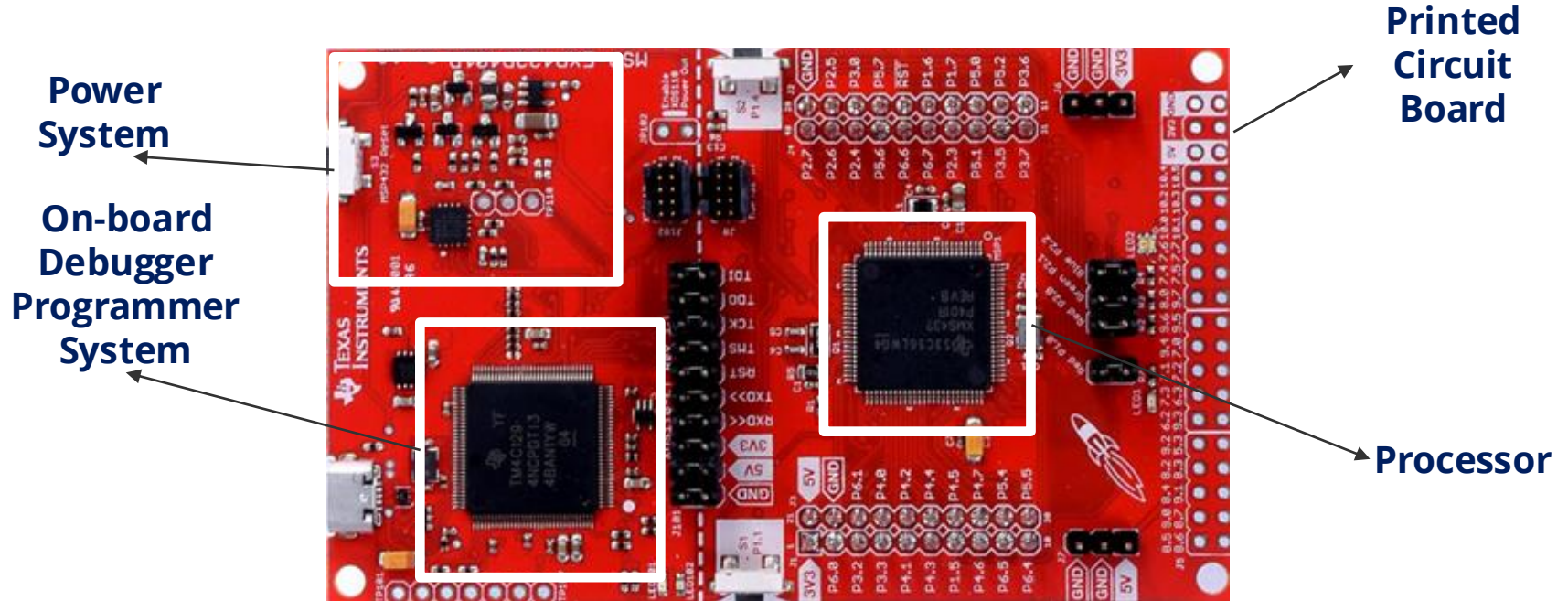
Embedded System Development Platform



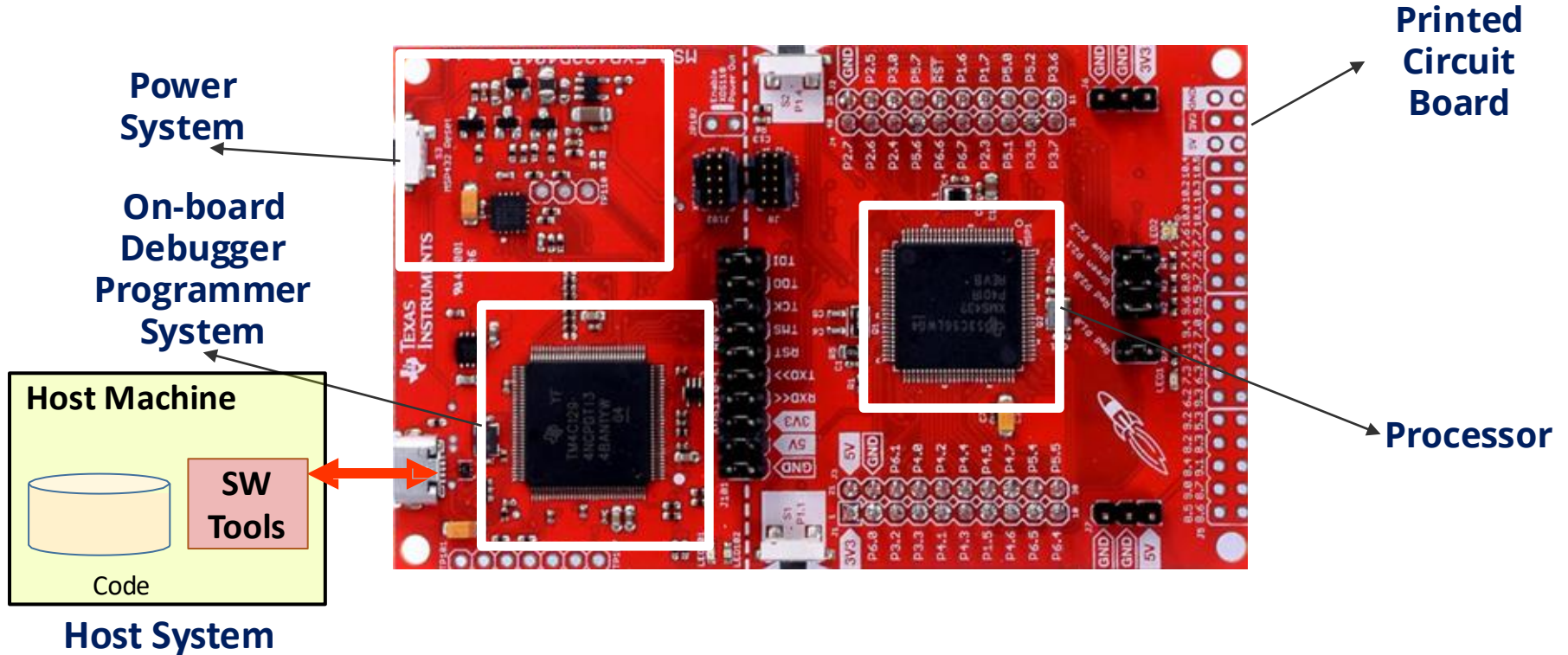
**Printed
Circuit
Board**

Processor

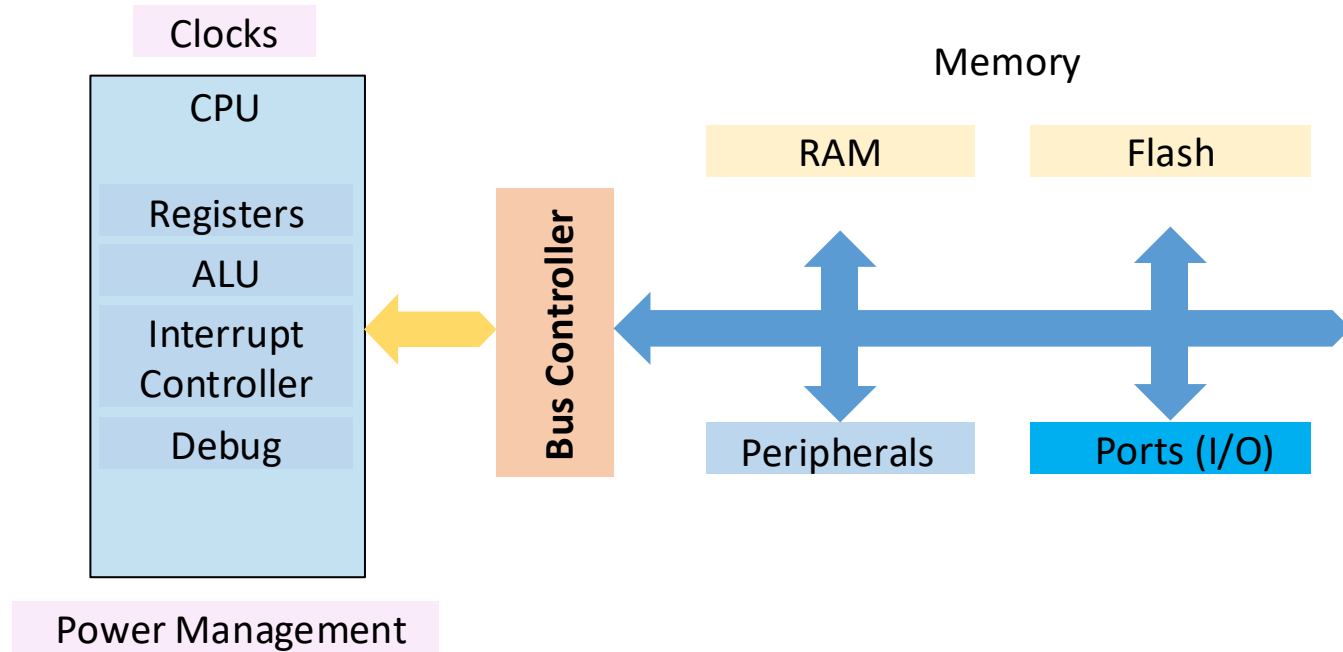
Embedded System Development Platform



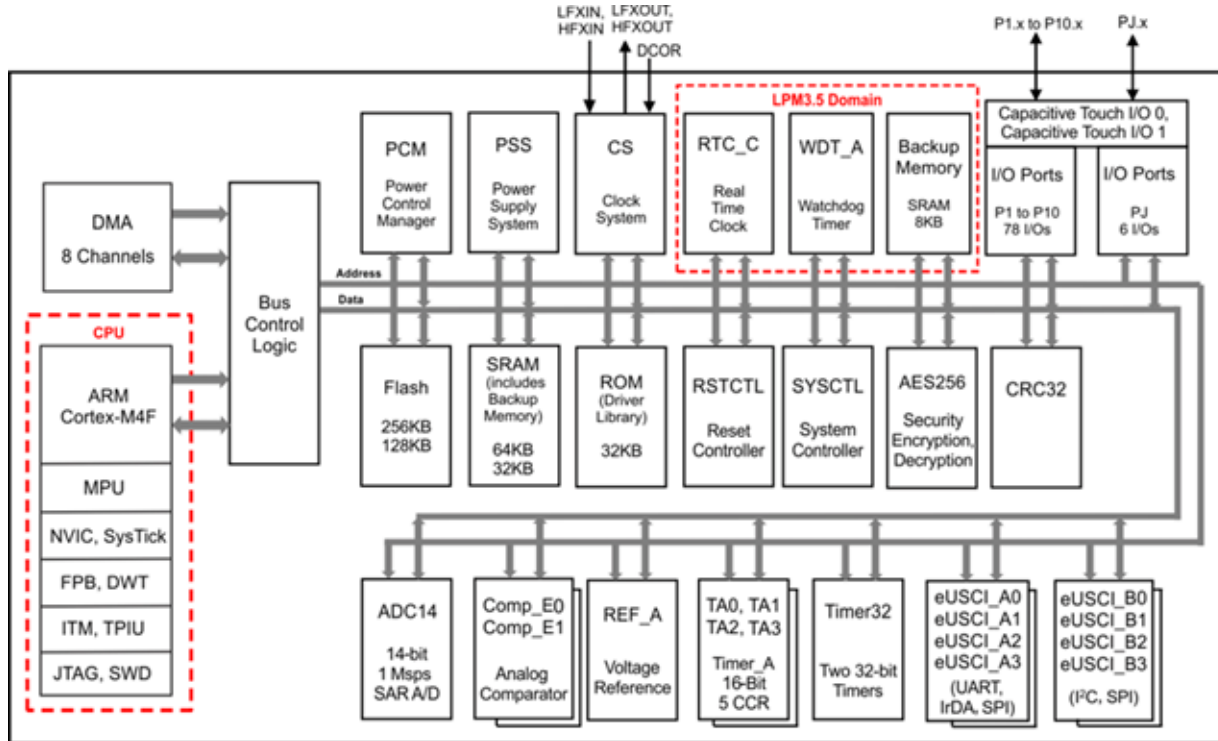
Embedded System Development Platform



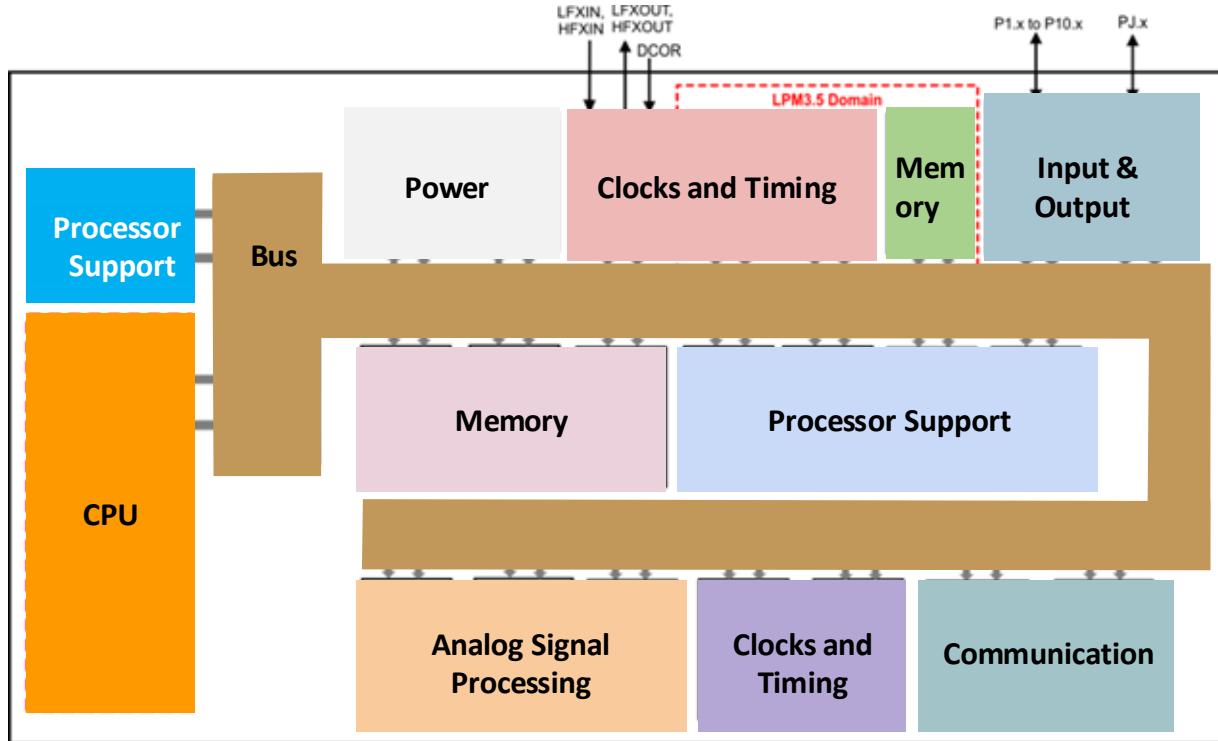
Microcontroller Components



MSP432 Cortex-M4 Microcontroller



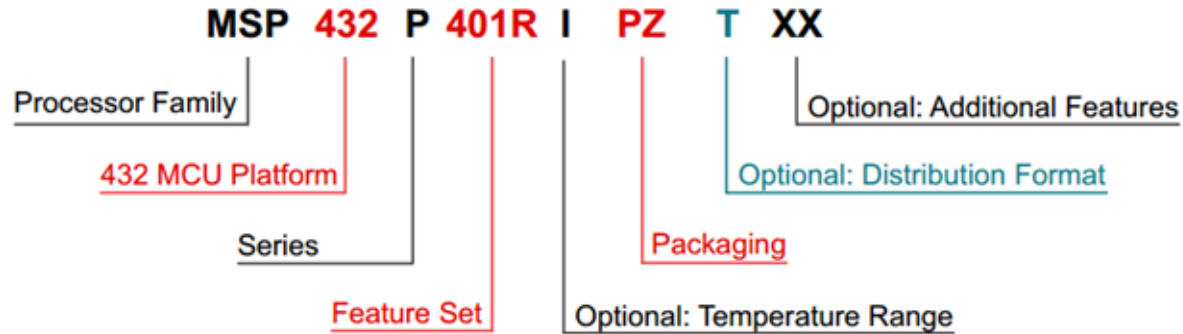
MSP432 Cortex-M4 Microcontroller



MSP432P401R Main Features

- ARM® Cortex™-M4F processor
- 100-pin microcontroller chip
- MSP stands for Mixed Signal Processor (analog plus digital)
- 256K bytes (256KB) of on-chip Flash memory for code
- 64KB of on-chip SRAM for data
- A large number of on-chip peripherals

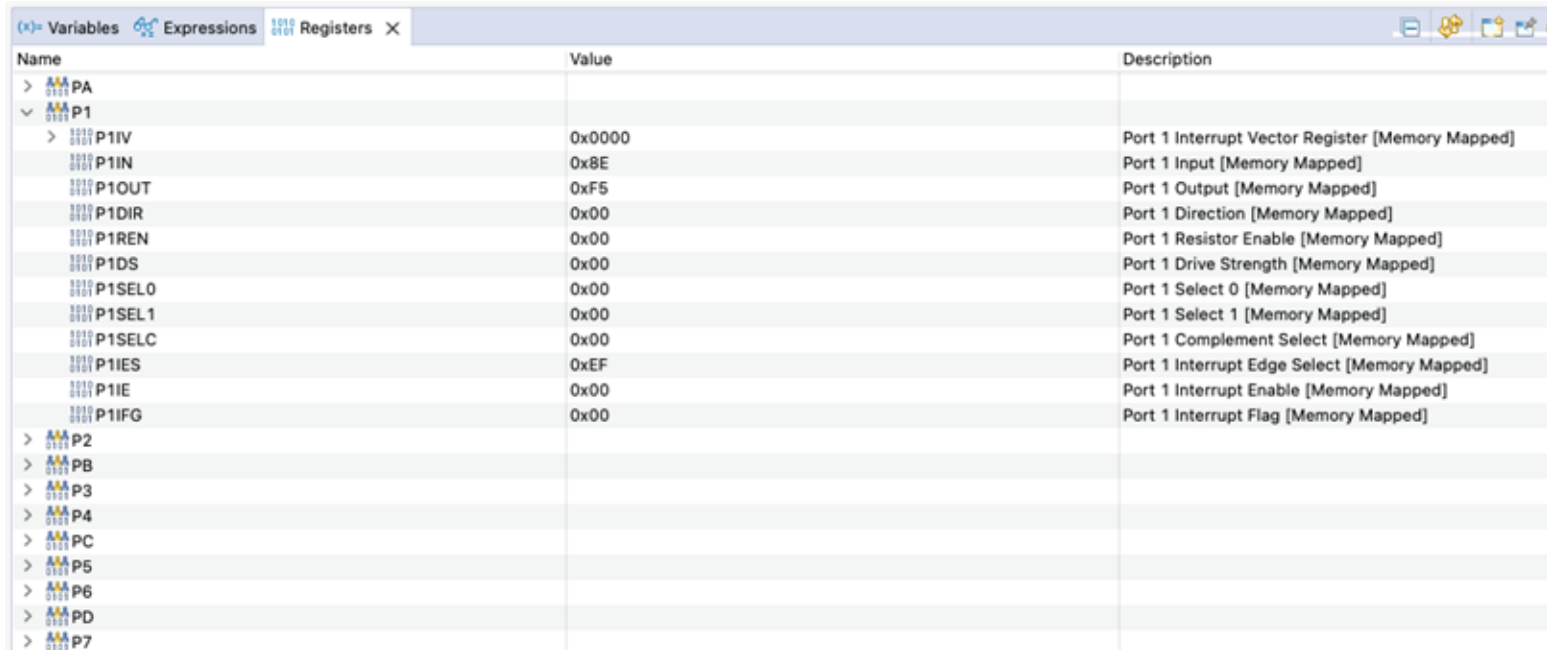
TI'S Naming Convention For MSP432



Processor Family	MSP = Mixed Signal Processor XMS = Experimental Silicon			
432 MCU Platform	TI's 32-bit Low-Power Microcontroller Platform			
Series	P = Performance and Low-Power Series			
Feature Set	First Digit 4 = Flash based devices up to 48 MHz	Second Digit 0 = General Purpose	Third Digit 1 = ADC14	Fourth Digit R = 256KB of Flash. 64KB of SRAM M = 128KB of Flash, 32KB of SRAM
Optional: Temperature Range	S = 0°C to 50 °C I = 40 °C to 85 °C T = -40 °C to 105 °C			
Packaging	PZ = LQFP			

TI Code Composer Studio (CCS)

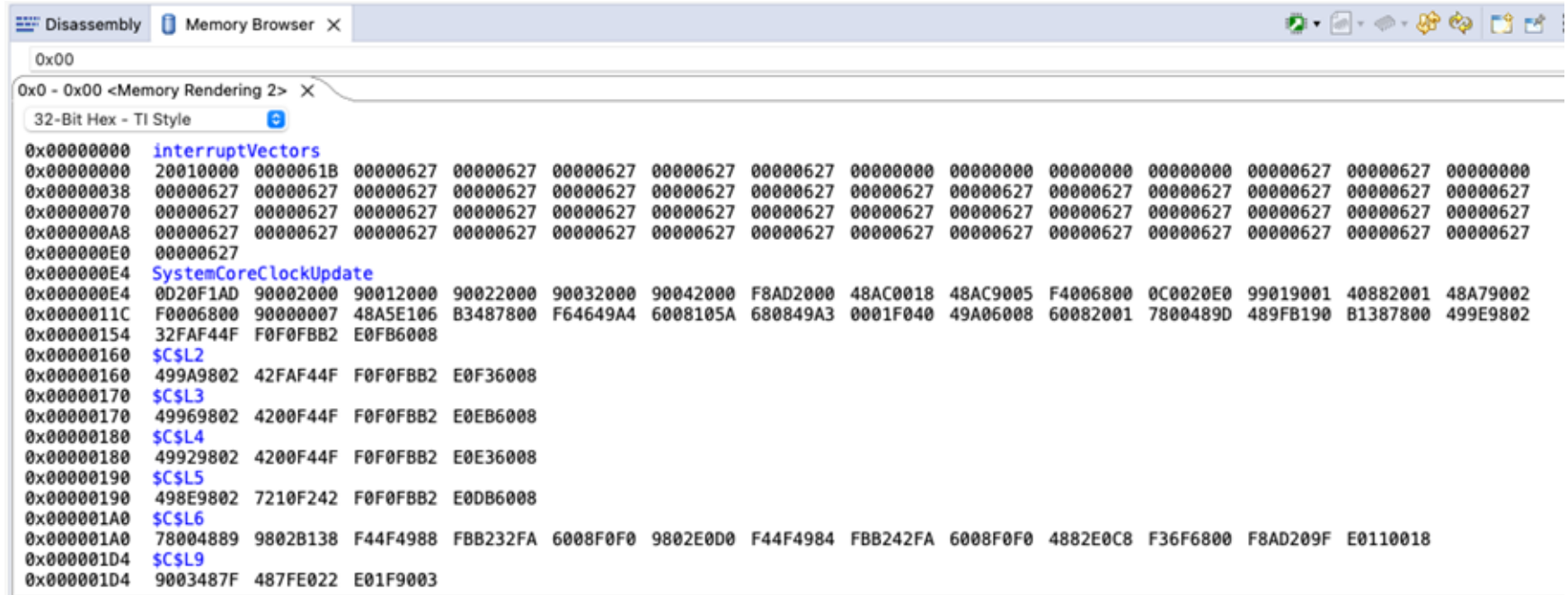
- Variables, Expressions, Registers



Name	Value	Description
> PA		
▼ P1		
> P1IV	0x0000	Port 1 Interrupt Vector Register [Memory Mapped]
P1IN	0x8E	Port 1 Input [Memory Mapped]
P1OUT	0xF5	Port 1 Output [Memory Mapped]
P1DIR	0x00	Port 1 Direction [Memory Mapped]
P1REN	0x00	Port 1 Resistor Enable [Memory Mapped]
P1DS	0x00	Port 1 Drive Strength [Memory Mapped]
P1SEL0	0x00	Port 1 Select 0 [Memory Mapped]
P1SEL1	0x00	Port 1 Select 1 [Memory Mapped]
P1SELC	0x00	Port 1 Complement Select [Memory Mapped]
P1IES	0xEF	Port 1 Interrupt Edge Select [Memory Mapped]
P1IE	0x00	Port 1 Interrupt Enable [Memory Mapped]
P1IFG	0x00	Port 1 Interrupt Flag [Memory Mapped]
> P2		
> PB		
> P3		
> P4		
> PC		
> P5		
> P6		
> PD		
> P7		

TI CCS

- Interrupt Vectors (0x0)



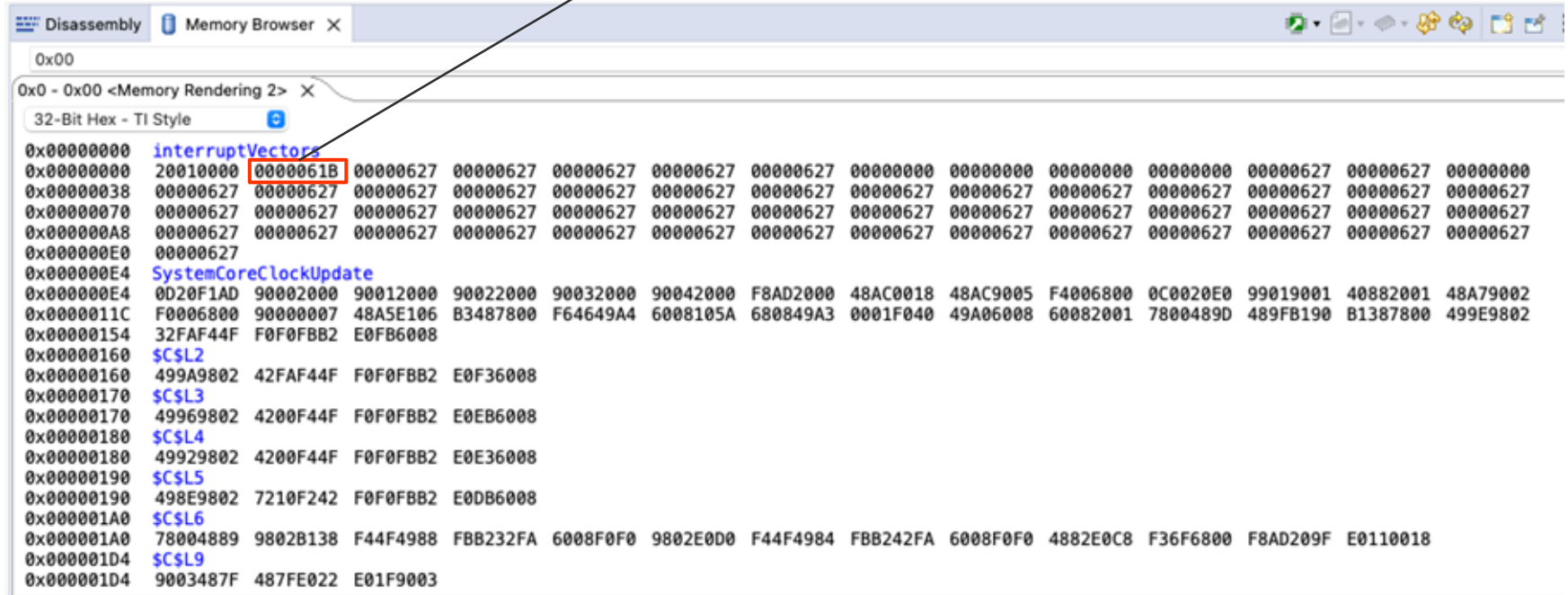
The screenshot shows the TI CCS Memory Browser window. The 'Disassembly' tab is selected, and the 'Memory Browser' sub-tab is active. The address range is set to '0x00'. The memory rendering is in '32-Bit Hex - TI Style'. The memory content is displayed in a table format, showing the Interrupt Vectors table. The table starts at address 0x00000000 and ends at 0x000001D4. The first entry is 'InterruptVectors' at 0x00000000, which points to 20010000. The second entry is 'SystemCoreClockUpdate' at 0x000000E4, which points to 0D20F1AD. The third entry is '\$CSL2' at 0x00000160, which points to 499A9802. The fourth entry is '\$CSL3' at 0x00000170, which points to 49969802. The fifth entry is '\$CSL4' at 0x00000180, which points to 49929802. The sixth entry is '\$CSL5' at 0x00000190, which points to 498E9802. The seventh entry is '\$CSL6' at 0x000001A0, which points to 78004889. The eighth entry is '\$CSL9' at 0x000001D4, which points to 9003487F.

Address	Label	Value
0x00000000	InterruptVectors	20010000
0x00000003		00000627
0x00000007		00000627
0x0000000A		00000627
0x0000000D		00000627
0x00000010		00000627
0x00000013		00000627
0x00000016		00000627
0x00000019		00000627
0x0000001C		00000627
0x0000001F		00000627
0x00000022		00000627
0x00000025		00000627
0x00000028		00000627
0x0000002B		00000627
0x0000002E		00000627
0x00000031		00000627
0x00000034		00000627
0x00000037		00000627
0x0000003A		00000627
0x0000003D		00000627
0x00000040		00000627
0x00000043		00000627
0x00000046		00000627
0x00000049		00000627
0x0000004C		00000627
0x0000004F		00000627
0x00000052		00000627
0x00000055		00000627
0x00000058		00000627
0x0000005B		00000627
0x0000005E		00000627
0x00000061		00000627
0x00000064		00000627
0x00000067		00000627
0x0000006A		00000627
0x0000006D		00000627
0x00000070		00000627
0x00000073		00000627
0x00000076		00000627
0x00000079		00000627
0x0000007C		00000627
0x0000007F		00000627
0x00000082		00000627
0x00000085		00000627
0x00000088		00000627
0x0000008B		00000627
0x0000008E		00000627
0x00000091		00000627
0x00000094		00000627
0x00000097		00000627
0x0000009A		00000627
0x0000009D		00000627
0x000000A0		00000627
0x000000A3		00000627
0x000000A6		00000627
0x000000A9		00000627
0x000000AC		00000627
0x000000AF		00000627
0x000000B2		00000627
0x000000B5		00000627
0x000000B8		00000627
0x000000BB		00000627
0x000000BE		00000627
0x000000C1		00000627
0x000000C4		00000627
0x000000C7		00000627
0x000000CA		00000627
0x000000CD		00000627
0x000000D0		00000627
0x000000D3		00000627
0x000000D6		00000627
0x000000D9		00000627
0x000000DC		00000627
0x000000DF		00000627
0x000000E2		00000627
0x000000E5		00000627
0x000000E8		00000627
0x000000EB		00000627
0x000000EE		00000627
0x000000F1		00000627
0x000000F4		00000627
0x000000F7		00000627
0x000000FA		00000627
0x000000FD		00000627
0x00000100		00000627
0x00000103		00000627
0x00000106		00000627
0x00000109		00000627
0x0000010C		00000627
0x0000010F		00000627
0x00000112		00000627
0x00000115		00000627
0x00000118		00000627
0x0000011B		00000627
0x0000011E		00000627
0x00000121		00000627
0x00000124		00000627
0x00000127		00000627
0x0000012A		00000627
0x0000012D		00000627
0x00000130		00000627
0x00000133		00000627
0x00000136		00000627
0x00000139		00000627
0x0000013C		00000627
0x0000013F		00000627
0x00000142		00000627
0x00000145		00000627
0x00000148		00000627
0x0000014B		00000627
0x0000014E		00000627
0x00000151		00000627
0x00000154		00000627
0x00000157		00000627
0x0000015A		00000627
0x0000015D		00000627
0x00000160		00000627
0x00000163		00000627
0x00000166		00000627
0x00000169		00000627
0x0000016C		00000627
0x0000016F		00000627
0x00000172		00000627
0x00000175		00000627
0x00000178		00000627
0x0000017B		00000627
0x0000017E		00000627
0x00000181		00000627
0x00000184		00000627
0x00000187		00000627
0x0000018A		00000627
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0x0000019C		00000627
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0x000001A5		00000627
0x000001A8		00000627
0x000001AB		00000627
0x000001AE		00000627
0x000001B1		00000627
0x000001B4		00000627
0x000001B7		00000627
0x000001BA		00000627
0x000001BD		00000627
0x000001C0		00000627
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0x000001CC		00000627
0x000001CF		00000627
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0x000001D5		00000627
0x000001D8		00000627
0x000001DB		00000627
0x000001DE		00000627
0x000001E1		00000627
0x000001E4		00000627
0x000001E7		00000627
0x000001EA		00000627
0x000001ED		00000627
0x000001F0		00000627
0x000001F3		00000627
0x000001F6		00000627
0x000001F9		00000627
0x000001FC		00000627
0x000001FF		00000627
0x00000200		00000627
0x00000203		00000627
0x00000206		00000627
0x00000209		00000627
0x0000020C		00000627
0x0000020F		00000627
0x00000212		00000627
0x00000215		00000627
0x00000218		00000627
0x0000021B		00000627
0x0000021E		00000627
0x00000221		00000627
0x00000224		00000627
0x00000227		00000627
0x0000022A		00000627
0x0000022D		00000627
0x00000230		00000627
0x00000233		00000627
0x00000236		00000627
0x00000239		00000627
0x0000023C		00000627
0x0000023F		00000627
0x00000242		00000627
0x00000245		00000627
0x00000248		00000627
0x0000024B		00000627
0x0000024E		00000627
0x00000251		00000627
0x00000254		00000627
0x00000257		00000627
0x0000025A		00000627
0x0000025D		00000627
0x00000260		00000627
0x00000263		00000627
0x00000266		00000627
0x00000269		00000627
0x0000026C		00000627
0x0000026F		00000627
0x00000272		00000627
0x00000275		00000627
0x00000278		00000627
0x0000027B		00000627
0x0000027E		00000627
0x00000281		00000627
0x00000284		00000627
0x00000287		00000627
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0x0000028D		00000627
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0x000002AB		00000627
0x000002AE		00000627
0x000002B1		00000627
0x000002B4		00000627
0x000002B7		00000627
0x000002BA		00000627
0x000002BD		00000627
0x000002C0		00000627
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0x000002D8		00000627
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0x000002DE		00000627
0x000002E1		00000627
0x000002E4		00000627
0x000002E7		00000627
0x000002EA		00000627
0x000002ED		00000627
0x000002F0		00000627
0x000002F3		00000627
0x000002F6		00000627
0x000002F9		00000627
0x000002FC		00000627
0x000002FF		00000627
0x00000300		00000627
0x00000303		00000627
0x00000306		00000627
0x00000309		00000627
0x0000030C		00000627
0x0000030F		00000627
0x00000312		00000627
0x00000315		00000627
0x00000318		00000627
0x0000031B		00000627
0x0000031E		00000627
0x00000321		00000627
0x00000324		00000627
0x00000327		00000627
0x0000032A		00000627
0x0000032D		00000627
0x00000330		00000627
0x00000333		00000627
0x00000336		00000627
0x00000339		00000627
0x0000033C		00000627
0x0000033F		00000627
0x00000342		00000627
0x00000345		00000627
0x00000348		00000627
0x0000034B		00000627
0x0000034E		00000627
0x00000351		00000627
0x00000354		00000627
0x00000357		00000627
0x0000035A		00000627
0x0000035D		00000627
0x00000360		00000627
0x00000363		00000627
0x00000366		00000627
0x00000369		00000627
0x0000036C		00000627
0x0000036F		00000627
0x00000372		00000627
0x00000375		00000627
0x00000378		00000627
0x0000037B		00000627
0x0000037E		00000627
0x00000381		00000627
0x00000384		00000627
0x00000387		00000627
0x0000038A		00000627
0x0000038D		00000627
0x00000390		00000627
0x00000393		00000627
0x00000396		00000627
0x00000399		00000627
0x0000039C		00000627
0x0000039F		00000627
0x000003A2		00000627
0x000003A5		00000627
0x000003A8		00000627
0x000003AB		00000627
0x000003AE		00000627
0x000003B1		00000627
0x000003B4		00000627
0x000003B7		00000627
0x000003BA		00000627
0x000003BD		00000627
0x000003C0		00000627
0x000003C3		00000627
0x000003C6		00000627
0x000003C9		00000627
0x000003CC		00000627
0x000003CF		00000627
0x000003D2		00000627
0x000003D5		00000627
0x000003D8</		

TI CCS

- Interrupt Vectors (0x0)

Reset
Vector



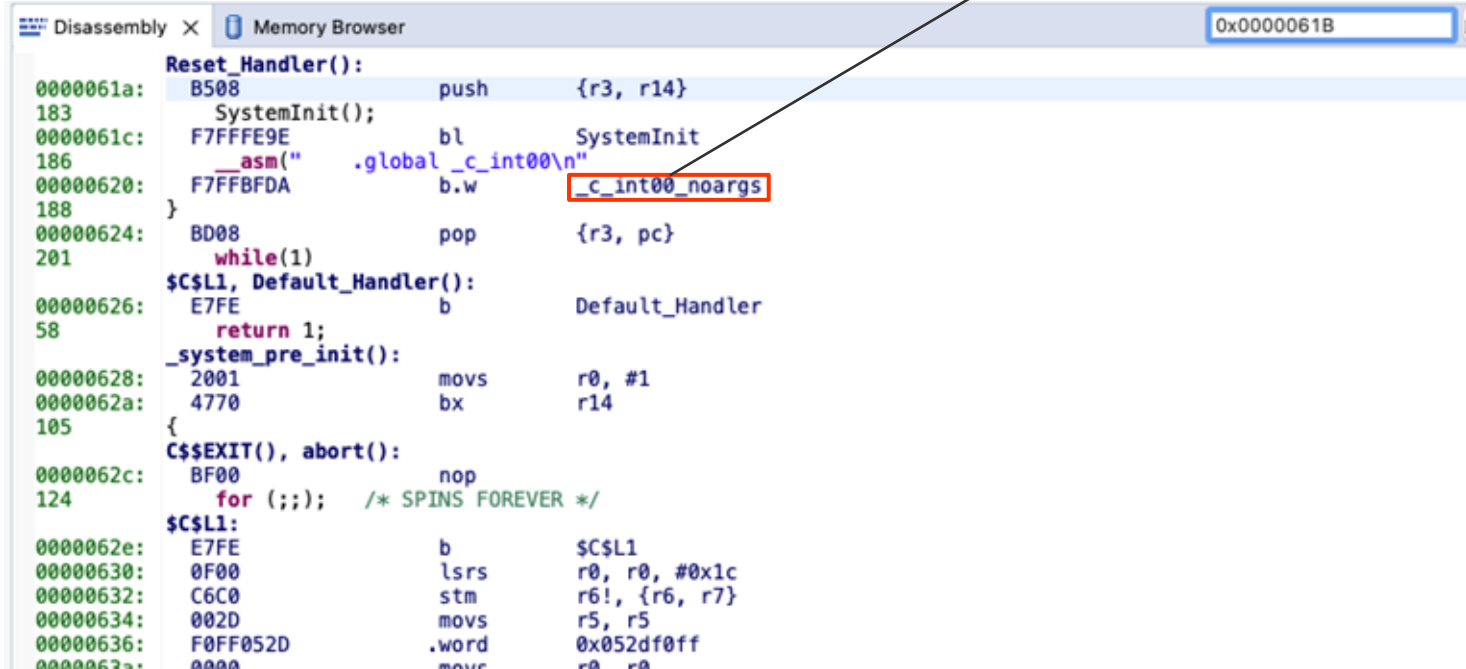
The screenshot shows the TI CCS Memory Browser window. The 'Memory Browser' tab is active, displaying the memory content for address 0x00. The '32-Bit Hex - TI Style' view is selected. The memory content is organized into sections: 'InterruptVectors' (starting at 0x00000000) and 'SystemCoreClockUpdate' (starting at 0x000000E4). The 'InterruptVectors' section contains a table of 16x16 values. The first row of this table, at address 0x00000000, contains the value 0000061B, which is highlighted with a red box. An arrow points from the 'Reset Vector' text to this value. The rest of the table contains the value 00000627. The 'SystemCoreClockUpdate' section contains a table of 16x16 values, with the first row at address 0x000000E4 containing various hexadecimal values.

Address	Value
0x00000000	0000061B
0x00000001	00000627
0x00000002	00000627
0x00000003	00000627
0x00000004	00000627
0x00000005	00000627
0x00000006	00000627
0x00000007	00000627
0x00000008	00000627
0x00000009	00000627
0x0000000A	00000627
0x0000000B	00000627
0x0000000C	00000627
0x0000000D	00000627
0x0000000E	00000627
0x0000000F	00000627
0x00000010	00000627
0x00000011	00000627
0x00000012	00000627
0x00000013	00000627
0x00000014	00000627
0x00000015	00000627
0x00000016	00000627
0x00000017	00000627
0x00000018	00000627
0x00000019	00000627
0x0000001A	00000627
0x0000001B	00000627
0x0000001C	00000627
0x0000001D	00000627
0x0000001E	00000627
0x0000001F	00000627
0x00000020	00000627
0x00000021	00000627
0x00000022	00000627
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0x00000024	00000627
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0x00000026	00000627
0x00000027	00000627
0x00000028	00000627
0x00000029	00000627
0x0000002A	00000627
0x0000002B	00000627
0x0000002C	00000627
0x0000002D	00000627
0x0000002E	00000627
0x0000002F	00000627
0x00000030	00000627
0x00000031	00000627
0x00000032	00000627
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0x00000038	00000627
0x00000039	00000627
0x0000003A	00000627
0x0000003B	00000627
0x0000003C	00000627
0x0000003D	00000627
0x0000003E	00000627
0x0000003F	00000627
0x00000040	00000627
0x00000041	00000627
0x00000042	00000627
0x00000043	00000627
0x00000044	00000627
0x00000045	00000627
0x00000046	00000627
0x00000047	00000627
0x00000048	00000627
0x00000049	00000627
0x0000004A	00000627
0x0000004B	00000627
0x0000004C	00000627
0x0000004D	00000627
0x0000004E	00000627
0x0000004F	00000627
0x00000050	00000627
0x00000051	00000627
0x00000052	00000627
0x00000053	00000627
0x00000054	00000627
0x00000055	00000627
0x00000056	00000627
0x00000057	00000627
0x00000058	00000627
0x00000059	00000627
0x0000005A	00000627
0x0000005B	00000627
0x0000005C	00000627
0x0000005D	00000627
0x0000005E	00000627
0x0000005F	00000627
0x00000060	00000627
0x00000061	00000627
0x00000062	00000627
0x00000063	00000627
0x00000064	00000627
0x00000065	00000627
0x00000066	00000627
0x00000067	00000627
0x00000068	00000627
0x00000069	00000627
0x0000006A	00000627
0x0000006B	00000627
0x0000006C	00000627
0x0000006D	00000627
0x0000006E	00000627
0x0000006F	00000627
0x00000070	00000627
0x00000071	00000627
0x00000072	00000627
0x00000073	00000627
0x00000074	00000627
0x00000075	00000627
0x00000076	00000627
0x00000077	00000627
0x00000078	00000627
0x00000079	00000627
0x0000007A	00000627
0x0000007B	00000627
0x0000007C	00000627
0x0000007D	00000627
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0x000000B9	00000627
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0x000000BC	00000627
0x000000BD	00000627
0x000000BE	00000627
0x000000BF	00000627
0x000000C0	00000627
0x000000C1	00000627
0x000000C2	00000627
0x000000C3	00000627
0x000000C4	00000627
0x000000C5	00000627
0x000000C6	00000627
0x000000C7	00000627
0x000000C8	00000627
0x000000C9	00000627
0x000000CA	00000627
0x000000CB	00000627
0x000000CC	00000627
0x000000CD	00000627
0x000000CE	00000627
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0x000000D1	00000627
0x000000D2	00000627
0x000000D3	00000627
0x000000D4	00000627
0x000000D5	00000627
0x000000D6	00000627
0x000000D7	00000627
0x000000D8	00000627
0x000000D9	00000627
0x000000DA	00000627
0x000000DB	00000627
0x000000DC	00000627
0x000000DD	00000627
0x000000DE	00000627
0x000000DF	00000627
0x000000E0	00000627
0x000000E1	00000627
0x000000E2	00000627
0x000000E3	00000627
0x000000E4	00000627
0x000000E5	00000627
0x000000E6	00000627
0x000000E7	00000627
0x000000E8	00000627
0x000000E9	00000627
0x000000EA	00000627
0x000000EB	00000627
0x000000EC	00000627
0x000000ED	00000627
0x000000EE	00000627
0x000000EF	00000627
0x000000F0	00000627
0x000000F1	00000627
0x000000F2	00000627
0x000000F3	00000627
0x000000F4	00000627
0x000000F5	00000627
0x000000F6	00000627
0x000000F7	00000627
0x000000F8	00000627
0x000000F9	00000627
0x000000FA	00000627
0x000000FB	00000627
0x000000FC	00000627
0x000000FD	00000627
0x000000FE	00000627
0x000000FF	00000627

TI CCS

- Reset Vector (0x0000061B)

Jump to
[_c_int00_noargs](#)

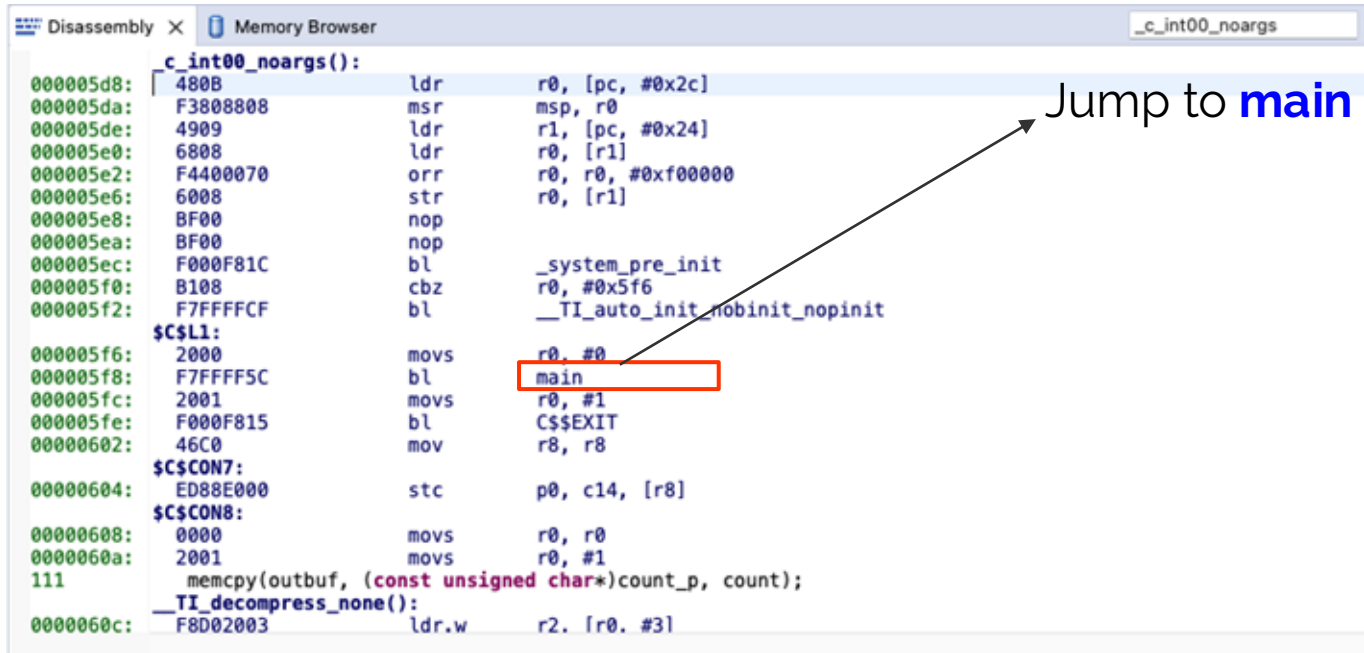


```
Disassembly X Memory Browser 0x0000061B

Reset_Handler():
0000061a: B508      push    {r3, r14}
183      SystemInit();
0000061c: F7FFFE9E  bl      SystemInit
186      _asm("    .global _c_int00\n"
00000620: F7FFBFDA  b.w     _c_int00_noargs
188      }
00000624: BD08      pop     {r3, pc}
201      while(1)
$C$L1, Default_Handler():
00000626: E7FE      b       Default_Handler
58      return 1;
_system_pre_init():
00000628: 2001      movs   r0, #1
0000062a: 4770      bx     r14
105      {
C$$EXIT(), abort():
0000062c: BF00      nop
124      for (;;); /* SPINS FOREVER */
$C$L1:
0000062e: E7FE      b       $C$L1
00000630: 0F00      lsrs   r0, r0, #0x1c
00000632: C6C0      stm    r6!, {r6, r7}
00000634: 002D      movs   r5, r5
00000636: F0FF052D .word   0x052df0ff
00000638: 0000      movs   r0, r0
```

TI CCS

- `_c_int00_noargs`

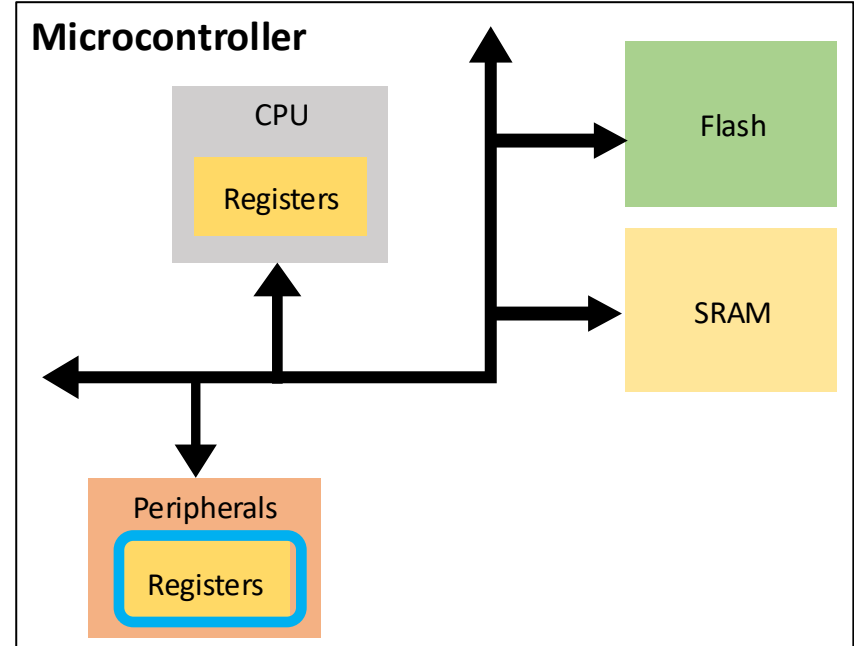


The screenshot shows the Disassembly window in TI CCS. The function `_c_int00_noargs()` is displayed, starting at address `000005d8`. The assembly code includes instructions for loading the program counter, setting the main stack pointer, and performing conditional branches. A red box highlights the `main` label in the instruction `bl main` at address `000005f8`. An arrow points from this label to the text "Jump to **main**".

```
Disassembly X Memory Browser _c_int00_noargs
_c_int00_noargs():
000005d8: 480B      ldr      r0, [pc, #0x2c]
000005da: F3808808  msr      msp, r0
000005de: 4909      ldr      r1, [pc, #0x24]
000005e0: 6808      ldr      r0, [r1]
000005e2: F4400070  orr      r0, r0, #0xf00000
000005e6: 6008      str      r0, [r1]
000005e8: BF00      nop
000005ea: BF00      nop
000005ec: F000F81C  bl       _system_pre_init
000005f0: B108      cbz      r0, #0x5f6
000005f2: F7FFFFCF  bl       __TI_auto_init_nobinit_nopinit
$CS$L1:
000005f6: 2000      movs     r0, #0
000005f8: F7FFFF5C  bl       main
000005fc: 2001      movs     r0, #1
000005fe: F000F815  bl       C$$EXIT
00000602: 46C0      mov      r8, r8
$CS$CON7:
00000604: ED88E000  stc      p0, c14, [r8]
$CS$CON8:
00000608: 0000      movs     r0, r0
0000060a: 2001      movs     r0, #1
111      memcpy(outbuf, (const unsigned char*)count_p, count);
__TI_decompress_none():
0000060c: F8D02003  ldr.w    r2, [r0, #3]
```

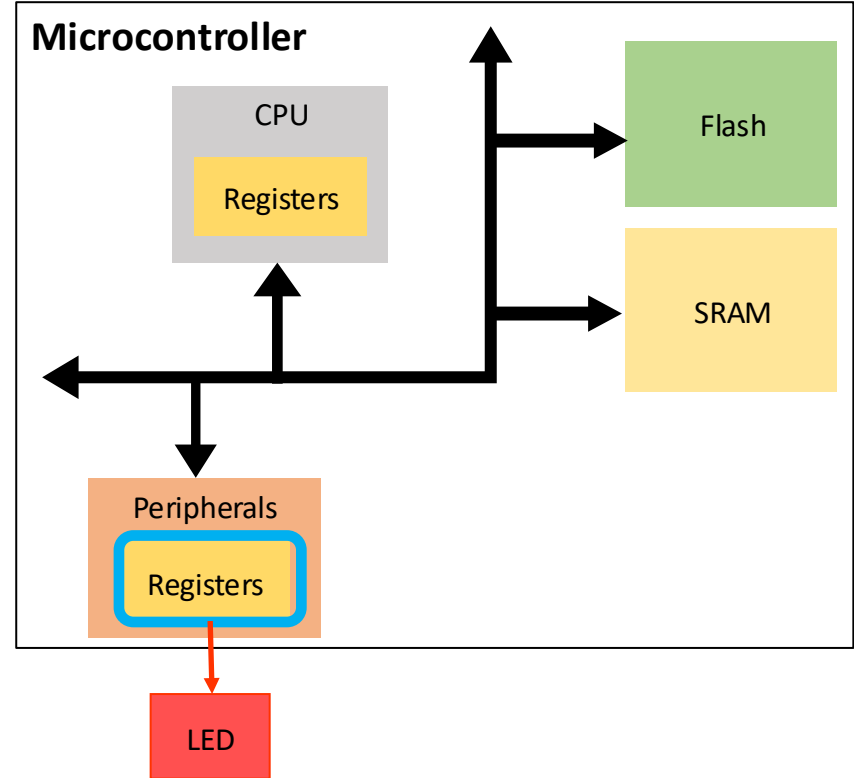
Peripherals inside MCU

- External to CPU but within the MCU.
 - **Memory-mapped I/O** to configure peripherals
 - Pointers
 - Memory Reads/Writes
 - Use of **Bit Manipulation**



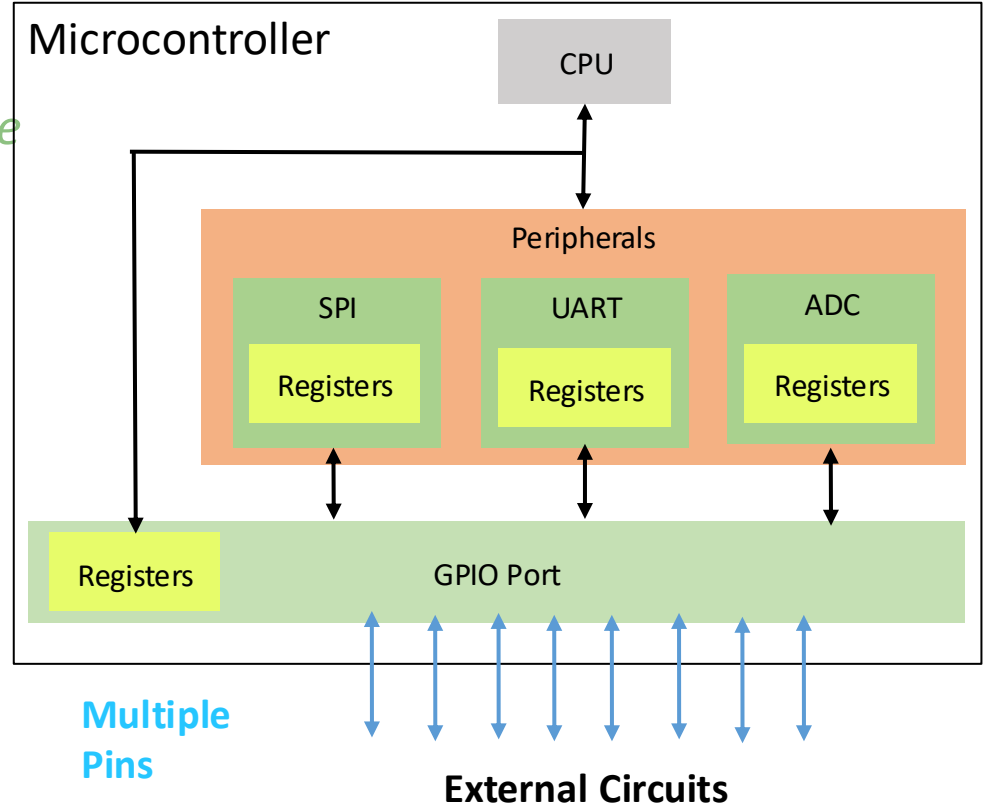
Peripherals inside MCU

- We use internal peripherals for several functions:
 - For instance to drive LEDs or to communicate with sensors.
- How to read inputs and/or provide outputs through peripherals?



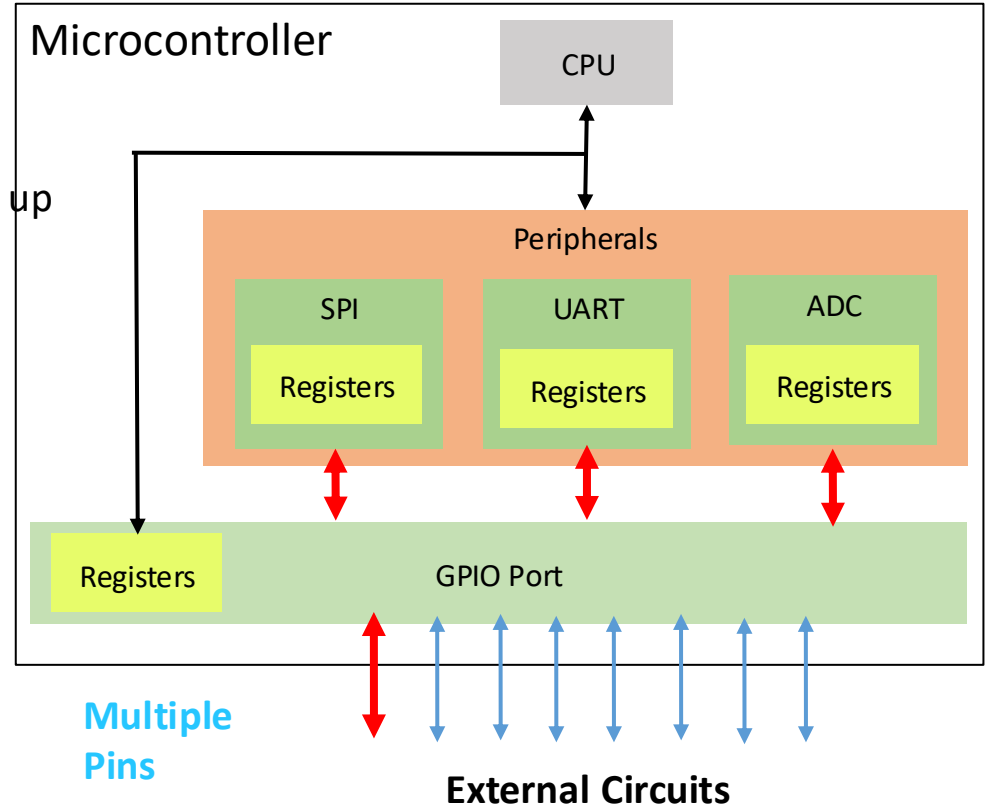
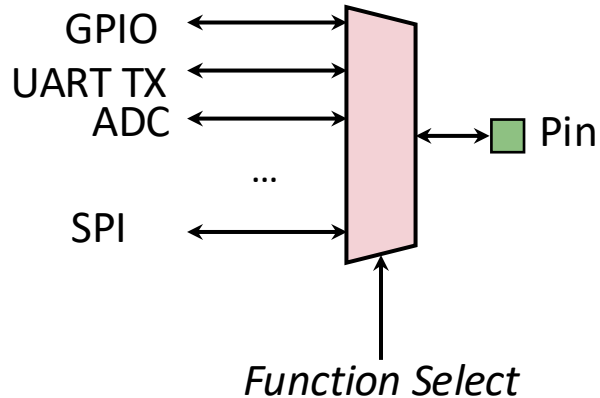
Input/Output Systems

- **GPIO peripheral** (*General Purpose Input/Output*)
 - **Pin** – physical connection to microcontroller
 - **Port** – combination of pins

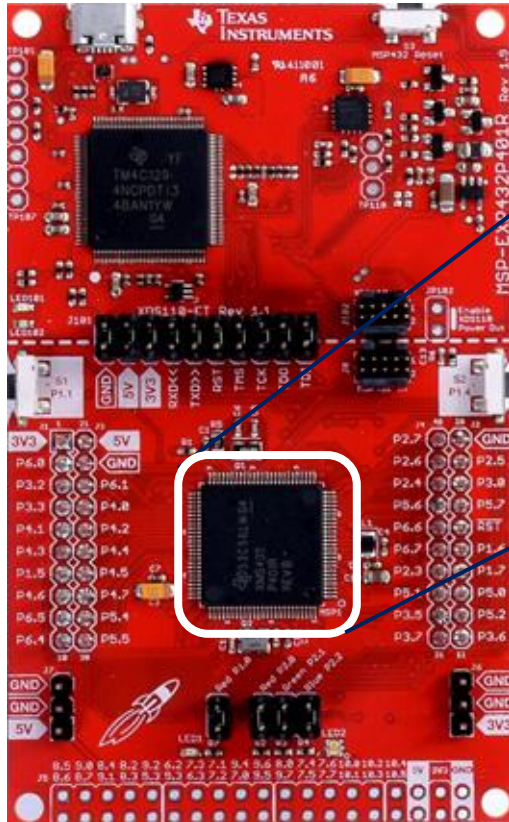


Input/Output Systems

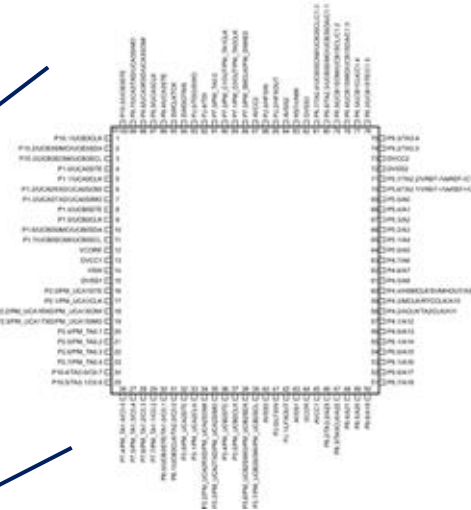
- Pins may have different features
 - To enable an alternative function, set up the appropriate register



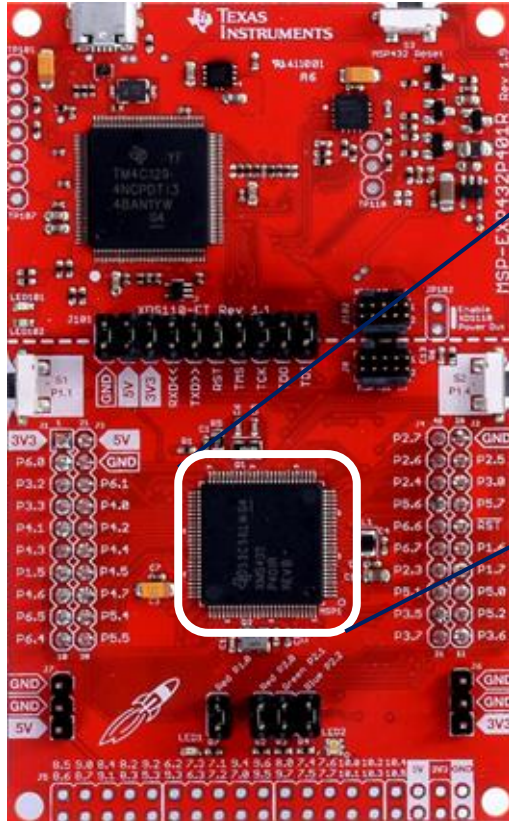
MSP432 Ports



10 Ports



MSP432 Ports



10 Ports



P1.0/UCA0STE	4
P1.1/UCA0CLK	5
P1.2/UCA0RXD/UCA0SOMI	6
P1.3/UCA0TXD/UCA0SIMO	7
P1.4/UCB0STE	8
P1.5/UCB0CLK	9
P1.6/UCB0SIMO/UCB0SDA	10
P1.7/UCB0SOMI/UCB0SCL	11

Port 1 – 8 Pins

GPIO / Primary / Secondary

Port 1 Pin 0 = P1.0

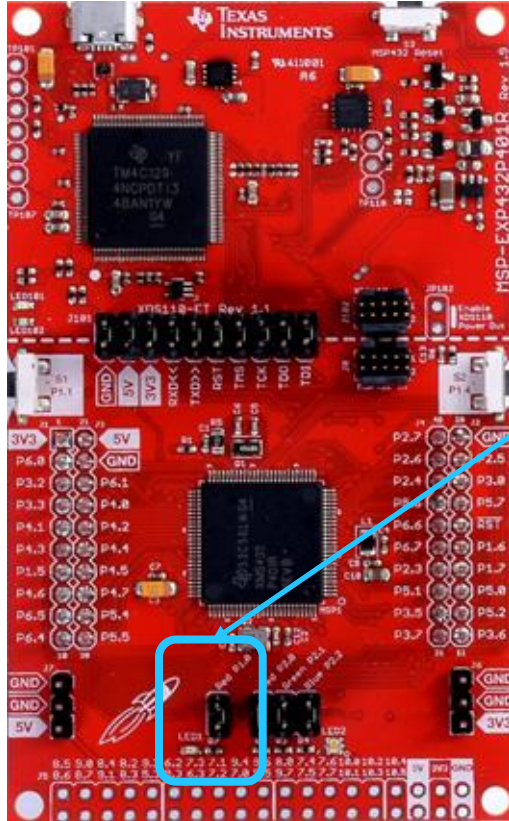
Primary Mode = SPI Chip Select

Port 1 Pin 7 = P1.7

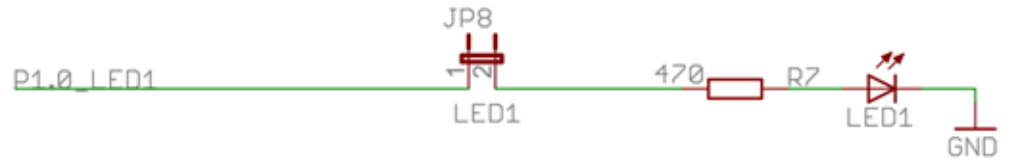
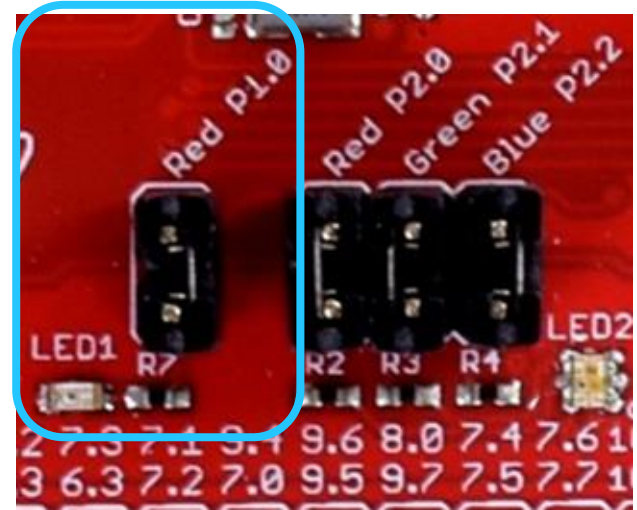
Primary Mode = SPI Slave-Out-Master-In Data (**UCB0SOMI**)

Secondary Mode = I²C Clock (**UCB0SCL**)

MSP432 Ports

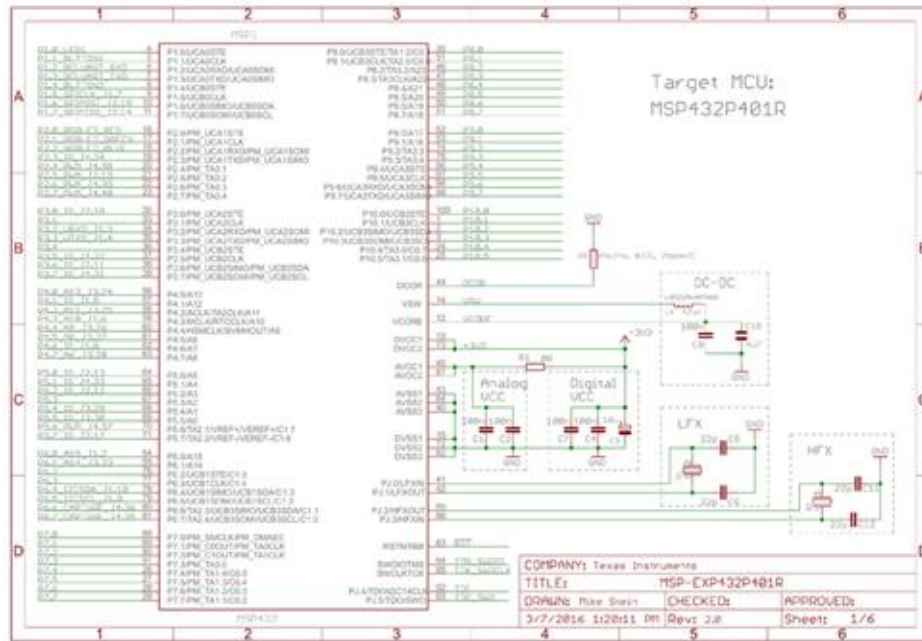


P1.0 Red LED Circuit



LED Schematic for P1.0

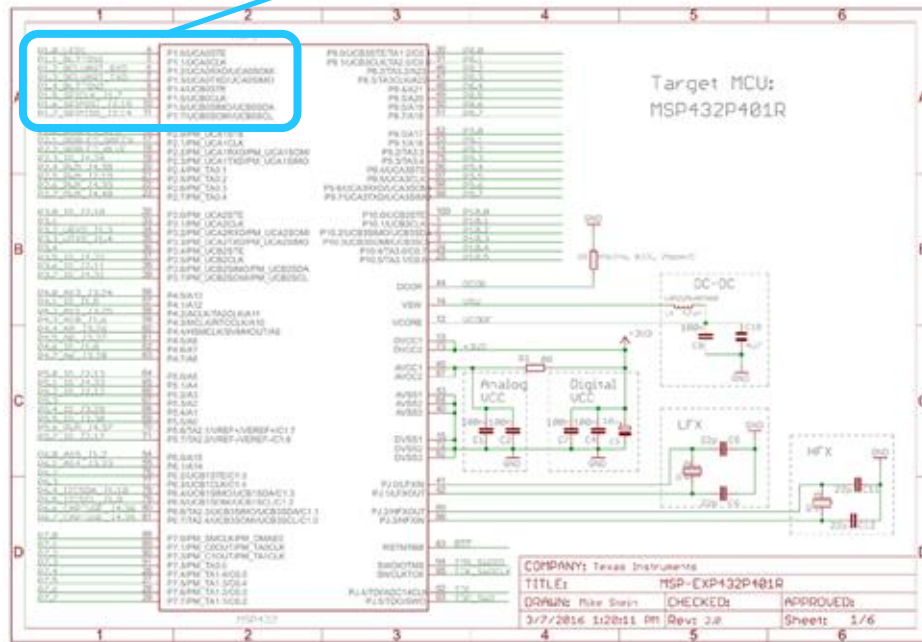
MSP432 Launchpad Schematic



Launchpad Schematic: MSP432 Pinout

MSP432 Launchpad Schematic

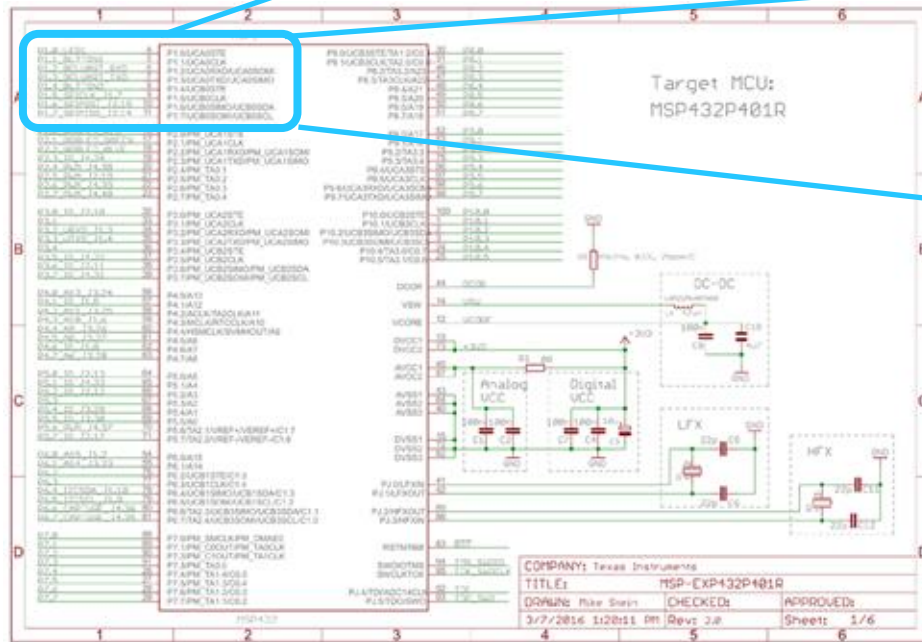
MSP432 Port 1



Launchpad Schematic: MSP432 Pinout

MSP432 Launchpad Schematic

MSP432 Port 1

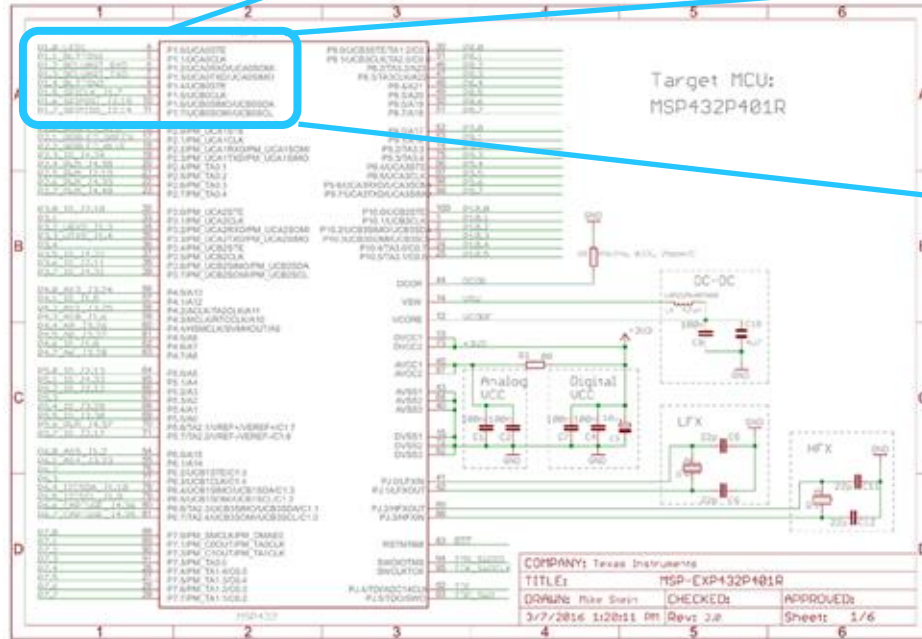


P1.0_LED1	4	P1.0/UCAB0STE
P1.1_BUTTON1	5	P1.1/UCAB0CLK
P1.2_BCUART_RXD	6	P1.2/UCAB0RXD/UCAB0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAB0TXD/UCAB0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIM0ST_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Launchpad Schematic: MSP432 Pinout

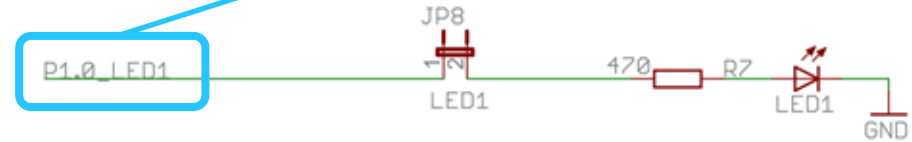
MSP432 Launchpad Schematic

MSP432 Port 1



P1.0_LED1	4	P1.0/UCAB0STE
P1.1_BUTTON1	5	P1.1/UCAB0CLK
P1.2_BCUART_RXD	6	P1.2/UCAB0RXD/UCAB0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAB0TXD/UCAB0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIM0ST_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Schematic Label for P1.0: P1.0_LED1

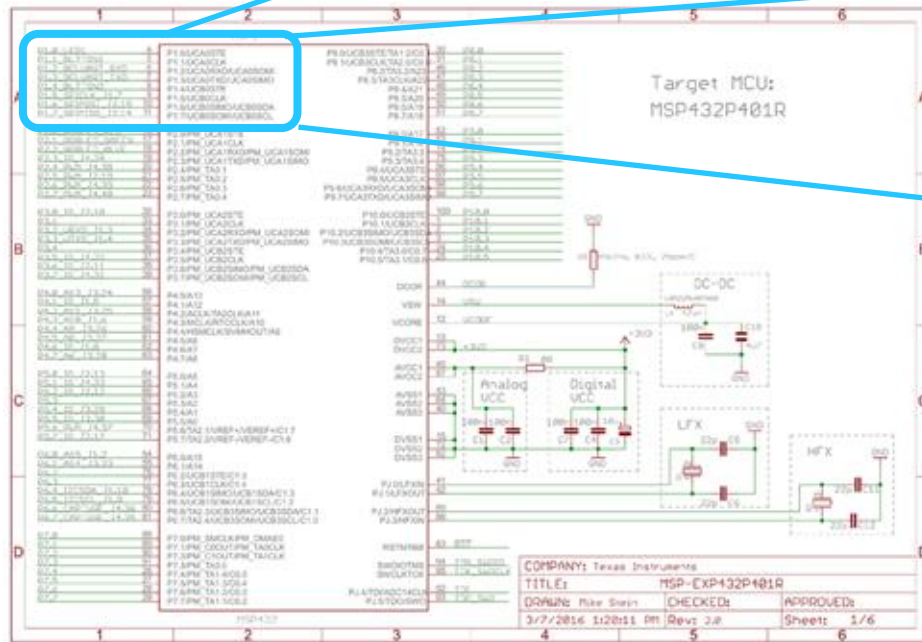


Launchpad Schematic: MSP432 Pinout

RED LED Schematic

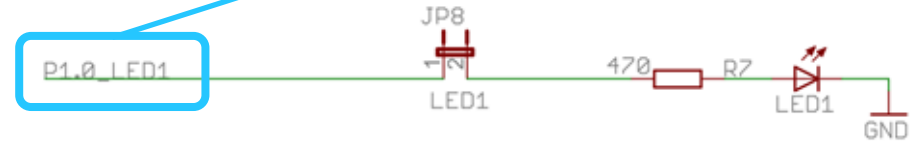
MSP432 Launchpad Schematic

MSP432 Port 1



P1.0_LED1	4	P1.0/UCAB0STE
P1.1_BUTTON1	5	P1.1/UCAB0CLK
P1.2_BCUART_RXD	6	P1.2/UCAB0RXD/UCAB0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAB0TXD/UCAB0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIM0ST_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Schematic Label for P1.0: **P1.0_LED1**

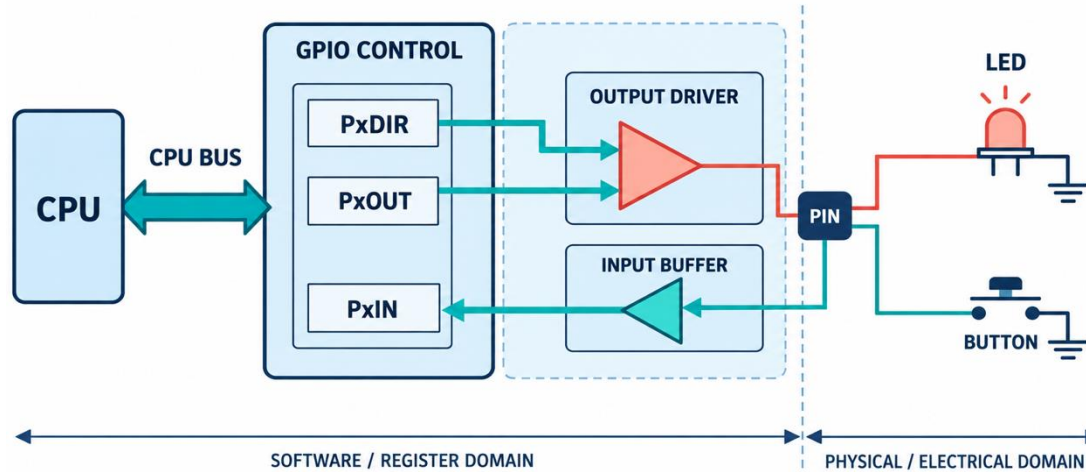


Launchpad Schematic: MSP432 Pinout

RED LED Schematic
How to control P1.0 in software?

GPIO Hardware Configuration

- High-level overview:
 - We have registers to control ports and pins.



GPIO Hardware Configuration

- To configure LED on P1.0

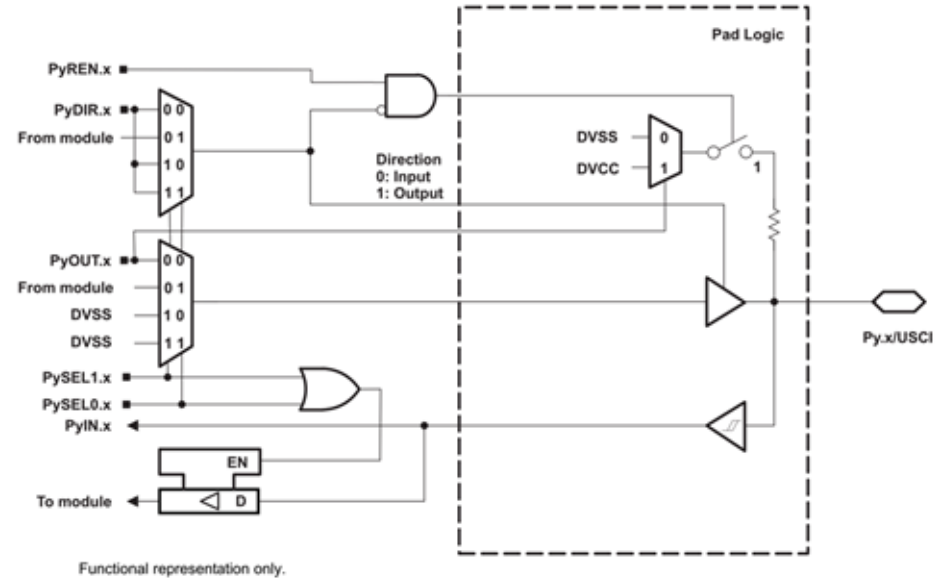
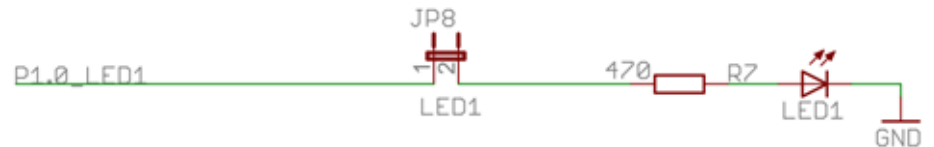


Figure 6-17. Py.x/USCI Pin Schematic



GPIO Hardware Configuration

- To configure LED on P1.0
 - Set **direction** to output
 - P1DIR

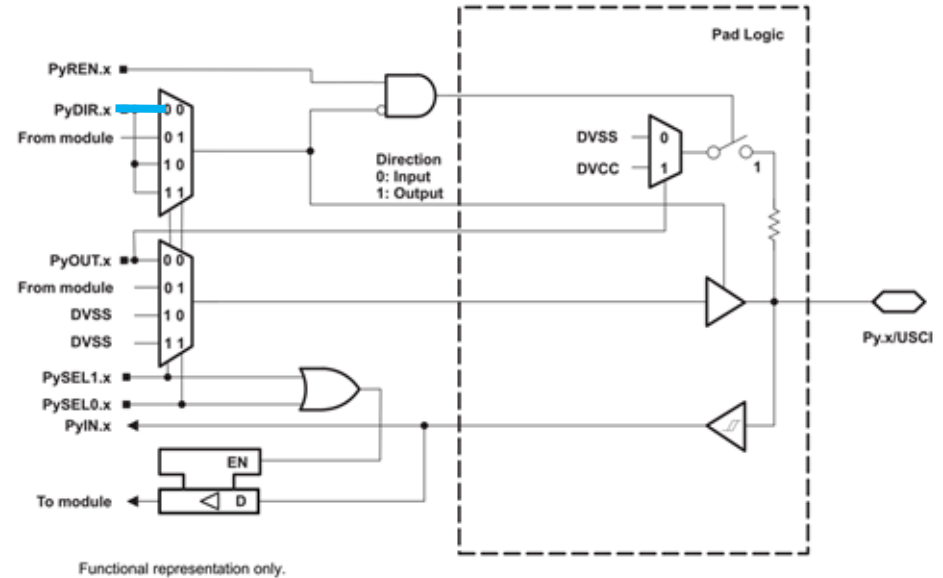
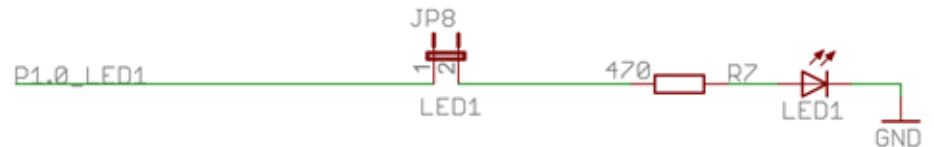


Figure 6-17. Py.x/USCI Pin Schematic



GPIO Hardware Configuration

- To configure LED on P1.0
 - Set **direction** to output
 - P1DIR
 - Set P1.0 **output** to high
 - P1OUT

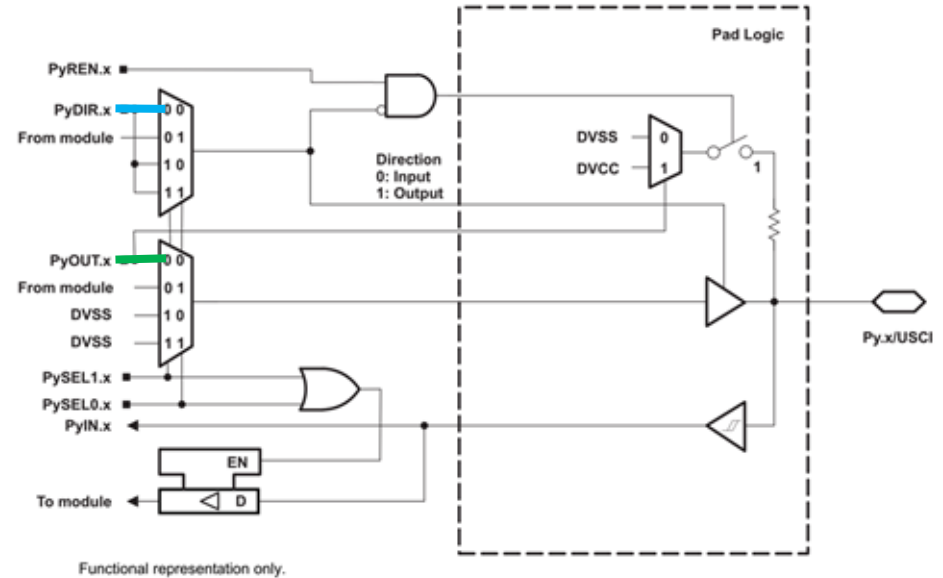
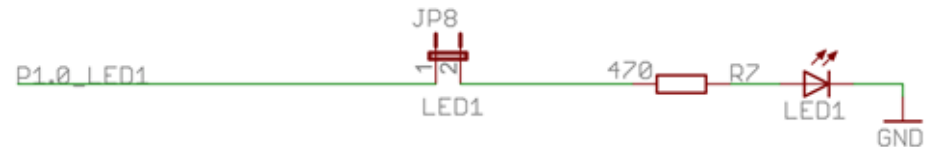


Figure 6-17. Py.x/USCI Pin Schematic



GPIO Hardware Configuration

- To configure LED on P1.0
 - Set **direction** to output
 - P1DIR
 - Set P1.0 **output** to high
 - P1OUT
 - Set P1SELx to **enable GPIO Mode**

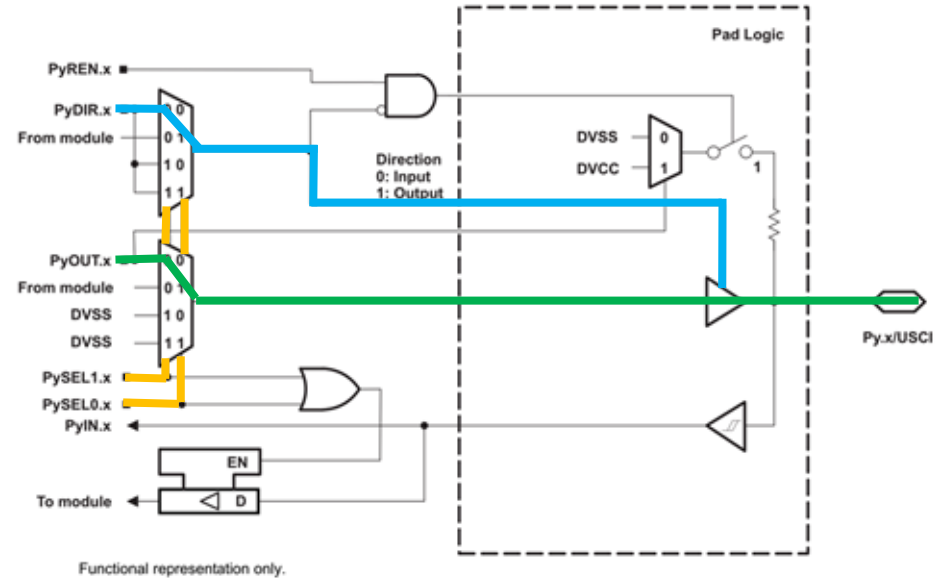
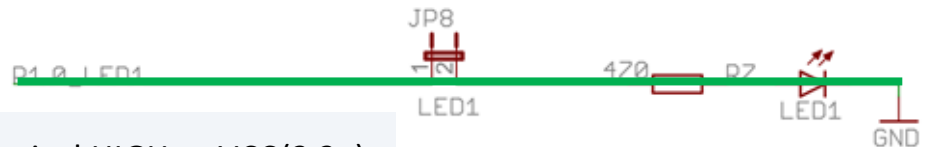


Figure 6-17. Py.x/USCI Pin Schematic



To turn on LED, P1.0_LED1 voltage needs to be a Logical HIGH -> VCC(3.3v)

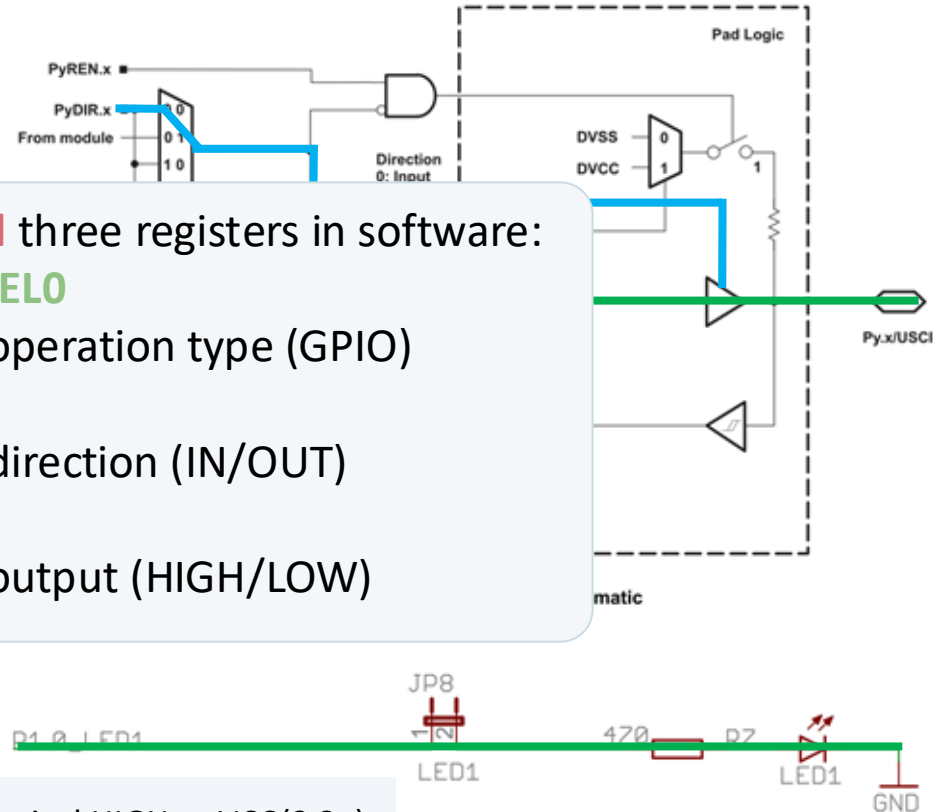
GPIO Hardware Configuration

- To configure LED on P1.0

- Set **direction** to output
 - P1DIR
- Set P1.0 **output** to HIGH
 - P1OUT
- Set P1SELx to **enable**

We need to **control** three registers in software:

- **P1SEL1** and **P1SEL0**
 - To set the operation type (GPIO)
- **P1DIR**
 - To set the direction (IN/OUT)
- **P1OUT**
 - To set the output (HIGH/LOW)



To turn on LED, P1.0_LED1 voltage needs to be a Logical HIGH -> VCC(3.3v)

MSP432 Launchpad Schematic

MSP432 Pinout Port 1

P1.0_LED1	4	P1.0/UCAS0STE
P1.1_BUTTON1	5	P1.1/UCAS0CLK
P1.2_BCUART_RXD	6	P1.2/UCAS0RXD/UCAS0SOMI
P1.3_BCUART_TXD	7	P1.3/UCAS0TXD/UCAS0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIMOSI_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

MSP432 Pin Descriptions

Table 6-32. Port P1 (P1.0 to P1.7) Pin Functions

PIN NAME (P1.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P1DIR.x	P1SEL1.x	P1SEL0.x
P1.0/UCAS0STE	0	P1.0 (I/O)	I: 0; O: 1	0	0
		UCAS0STE	X ⁽²⁾	0	1
		N/A	0	1	0
		DVSS	1		
		N/A	0	1	1
		DVSS	1		
P1.1/UCAS0CLK	1	P1.1 (I/O)	I: 0; O: 1	0	0
		UCAS0CLK	X ⁽²⁾	0	1
		N/A	0	1	0
		DVSS	1		
		N/A	0	1	1
		DVSS	1		
P1.2/UCAS0RXD/UCAS0SOMI	2	P1.2 (I/O)	I: 0; O: 1	0	0
		UCAS0RXD/UCAS0SOMI	X ⁽²⁾	0	1
		N/A	0	1	0
		DVSS	1		
		N/A	0	1	1
		DVSS	1		

MSP432 Launchpad Schematic

MSP432 Pinout Port 1

P1.0_LED1	4	P1.0/UCxA0STE
P1.1_BUTTON1	5	P1.1/UCxA0CLK
P1.2_BCLUART_RXD	6	P1.2/UCxA0RXD/UCxA0SOMI
P1.3_BCLUART_TXD	7	P1.3/UCxA0TXD/UCxA0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIMOSI_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Pin
Number

MSP432 Pin Descriptions

Table 6-32. Port P1 (P1.0 to P1.7) Pin Functions

PIN NAME (P1.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P1DIR.x	P1SEL1.x	P1SEL0.x
P1.0/UCxA0STE	0	P1.0 (I/O)	I: 0; O: 1	0	0
		UCxA0STE	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.1/UCxA0CLK	1	P1.1 (I/O)	I: 0; O: 1	0	0
		UCxA0CLK	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.2/UCxA0RXD/UCxA0SOMI	2	P1.2 (I/O)	I: 0; O: 1	0	0
		UCxA0RXD/UCxA0SOMI	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1

Port P1Sel0.x / P1Sel1.x registers
Select Mode for each Pin in Port

MSP432 Launchpad Schematic

MSP432 Pinout Port 1

P1.0_LED1	4	P1.0/UCxA0STE
P1.1_BUTTON1	5	P1.1/UCxA0CLK
P1.2_BCLUART_RXD	6	P1.2/UCxA0RXD/UCxA0SOMI
P1.3_BCLUART_TXD	7	P1.3/UCxA0TXD/UCxA0SIMO
P1.4_BUTTON2	8	P1.4/UCB0STE
P1.5_SPICLK_J1.7	9	P1.5/UCB0CLK
P1.6_SPIMOSI_J2.15	10	P1.6/UCB0SIMO/UCB0SDA
P1.7_SPIMISO_J2.14	11	P1.7/UCB0SOMI/UCB0SCL

Pin
Number

Pin
Modes

MSP432 Pin Descriptions

Table 6-32. Port P1 (P1.0 to P1.7) Pin Functions

PIN NAME (P1.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P1DIR.x	P1SEL1.x	P1SEL0.x
P1.0/UCxA0STE	0	P1.0 (I/O)	I: 0; O: 1	0	0
		UCxA0STE	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.1/UCxA0CLK	1	P1.1 (I/O)	I: 0; O: 1	0	0
		UCxA0CLK	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1
P1.2/UCxA0RXD/UCxA0SOMI	2	P1.2 (I/O)	I: 0; O: 1	0	0
		UCxA0RXD/UCxA0SOMI	X ⁽²⁾	0	1
		N/A	0		
		DVSS	1	1	0
		N/A	0		
		DVSS	1	1	1

Port P1Sel0.x / P1Sel1.x registers
Select Mode for each Pin in Port

Configuring LED

- Two important registers
 - IO Direction Register (P1DIR)
 - IO Output Register (P1OUT)

Table 12-3. Digital I/O Registers

Offset	Acronym	Register Name	Section
0Eh	P1IV	Port 1 Interrupt Vector	Section 12.4.1
1Eh	P2IV	Port 2 Interrupt Vector	Section 12.4.1
2Eh	P3IV	Port 3 Interrupt Vector	Section 12.4.1
3Eh	P4IV	Port 4 Interrupt Vector	Section 12.4.1
4Eh	P5IV	Port 5 Interrupt Vector	Section 12.4.1
5Eh	P6IV	Port 6 Interrupt Vector	Section 12.4.1
6Eh	P7IV	Port 7 Interrupt Vector	Section 12.4.1
7Eh	P8IV	Port 8 Interrupt Vector	Section 12.4.1
8Eh	P9IV	Port 9 Interrupt Vector	Section 12.4.1
9Eh	P10IV	Port 10 Interrupt Vector	Section 12.4.1
00h	P1IN or PAIN_L	Port 1 Input	Section 12.4.2
02h	P1OUT or PAOUT_L	Port 1 Output	Section 12.4.3
04h	P1DIR or PADIR_L	Port 1 Direction	Section 12.4.4
06h	P1REN or PAREN_L	Port 1 Resistor Enable	Section 12.4.5
08h	P1DS or PADS_L	Port 1 Drive Strength	Section 12.4.6
0Ah	P1SEL0 or PASEL0_L	Port 1 Select 0	Section 12.4.7
0Ch	P1SEL1 or PASEL1_L	Port 1 Select 1	Section 12.4.8
16h	P1SELC or PASELC_L	Port 1 Complement Selection	Section 12.4.9
18h	P1IES or PAIES_L	Port 1 Interrupt Edge Select	Section 12.4.10
1Ah	P1IE or PAIE_L	Port 1 Interrupt Enable	Section 12.4.11
1Ch	P1IFG or PAIFG_L	Port 1 Interrupt Flag	Section 12.4.12

Configuring LED

- Two important registers
 - IO Direction Register (P1DIR)
 - IO Output Register (P1OUT)
- PxOUT, PxDIR
 - x = Port Number
 - Examples: P2OUT, P1DIR

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

7	6	5	4	3	2	1	0
PxOUT							
RW	RW	RW	RW	RW	RW	RW	RW

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

12.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
RW-0	RW-0	RW-0	RW-0	RW-0	RW-0	RW-0	RW-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

P1DIR Register

- Port Direction Register sets the pin to:
 - 0: an input
 - 1: an output

```
/* Configure P1.0 output */  
P1->DIR |= BIT0;
```

rw-0: Bit is Read/Write & defaults to 0

12.4.4 PxDIR Register

Port X Direction Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-4. PxDIR Register

7	6	5	4	3	2	1	0
PxDIR							
rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0	rw-0

Table 12-7. PxDIR Register Description

Bit	Field	Type	Reset	Description
7-0	PxDIR	RW	0h	Port X direction. 0b = Port configured as input 1b = Port configured as output

P1OUT Register

- Port Output Register sets output to:
 - 0: Output set to LOW (GND)
 - 1: Output set to HIGH (VCC)

```
/* Configure P1.0 output */  
P1->DIR |= BIT0;  
  
/* Set P1.0 HIGH */  
P1->OUT |= BIT0;
```

12.4.3 PxOUT Register

Port X Output Register (X = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or J)

Figure 12-3. PxOUT Register

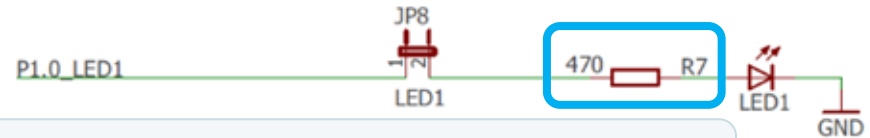
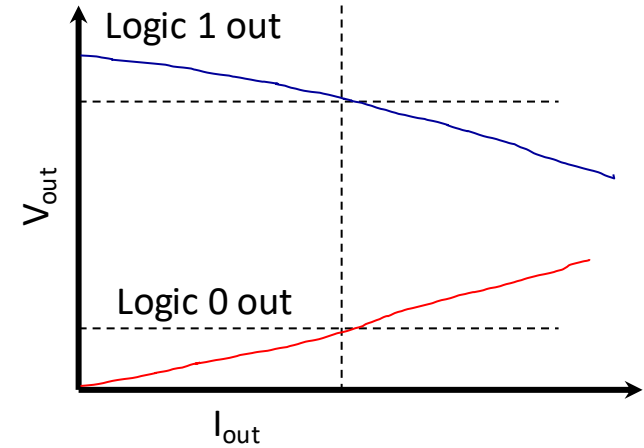
7	6	5	4	3	2	1	0
PxOUT							
rw	rw	rw	rw	rw	rw	rw	rw

Table 12-6. PxOUT Register Description

Bit	Field	Type	Reset	Description
7-0	PxOUT	RW	Undefined	Port X output. When I/O configured to output mode: 0b = Output is low. 1b = Output is high. When I/O configured to input mode and pullups/pulldowns enabled: 0b = Pulldown selected 1b = Pullup selected

I/O: What's a One? A Zero?

- The input signal's value is determined by the voltage
 - **Exceeding** VDD or GND **may damage the chip**
- Nominal output voltages
 - 1: VDD-0.5V to VDD, 0: 0V to 0.5V
 - **VDD = 3.3V** on MSP432



Limit current to a value that is safe for both LED and MCU port driver

Use **current-limiting resistor**: $R = (V_{DD} - V_{LED}) / I_{LED}$

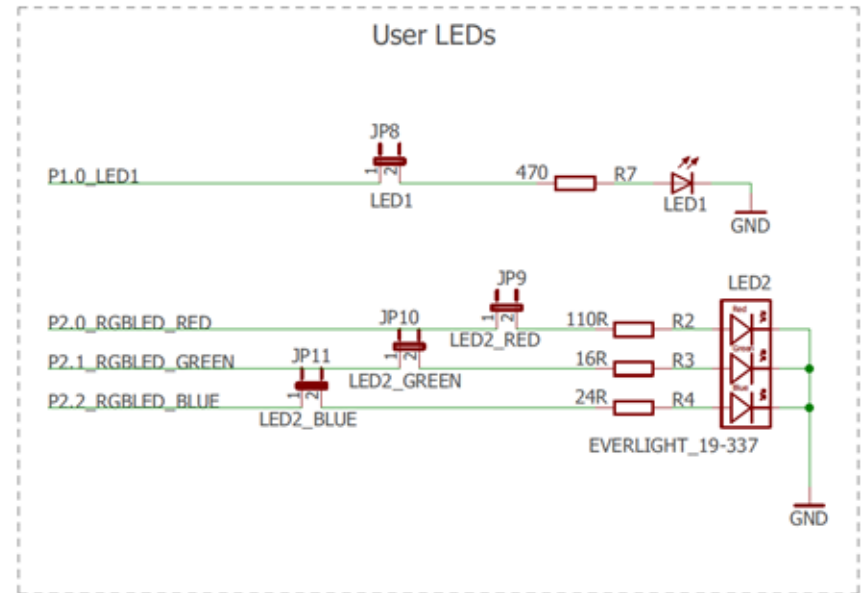
$I_{LED} \rightarrow$ Max. LED current

Solve for R given VDD = ~3.0V

Blinking Multiple LEDs

- Which ports?
- Which registers?
 - PxDIR?
 - PxOUT?

Buttons and LEDs



Exercise

- Install Software and try **Blink** examples.
- TI CCS: [Download here](#)
 - Download version **12.8.1 (Aug 9, 2024)!**